



The genome editing revolution

Andrew Bassett

RNA-programmed genome editing in human cells

Martin Jinek^{1,2}, Alexandra East², Aaron Cheng², Steven Lin^{1,2}, Enbo Ma², Jennifer Doudna^{1,2,3,4*}

¹Howard Hughes Medical Institute, University of California, Berkeley, Berkeley, United States; ²Department of Molecular and Cell Biology, University of California, Berkeley, Berkeley, United States; ³Department of Chemistry, University of California, Berkeley, Berkeley, United States; ⁴Physical Biosciences Division, Lawrence Berkeley National Laboratory, Berkeley, United States

Science*express*

RNA-Guided Human Genome Engineering via Cas9

Prashant Mali,^{1,5} Luhan Yang,^{1,3,5} Kevin M. Esvelt,² John Aach,¹ Marc Guell,¹ James E. DiCarlo,⁴ Julie E. Norville,¹ George M. Church^{1,2*}

Science*express*

Multiplex Genome Engineering Using CRISPR/Cas Systems

Le Cong,^{1,2*} F. Ann Ran,^{1,4*} David Cox,^{1,3} Shuailiang Lin,^{1,5} Robert Barretto,⁶ Naomi Habib,¹ Patrick D. Hsu,^{1,4} Xuebing Wu,⁷ Wenyang Jiang,⁸ Luciano A. Marraffini,⁶ Feng Zhang^{1†}

nature
biotechnology

Efficient genome editing in zebrafish using a CRISPR-Cas system

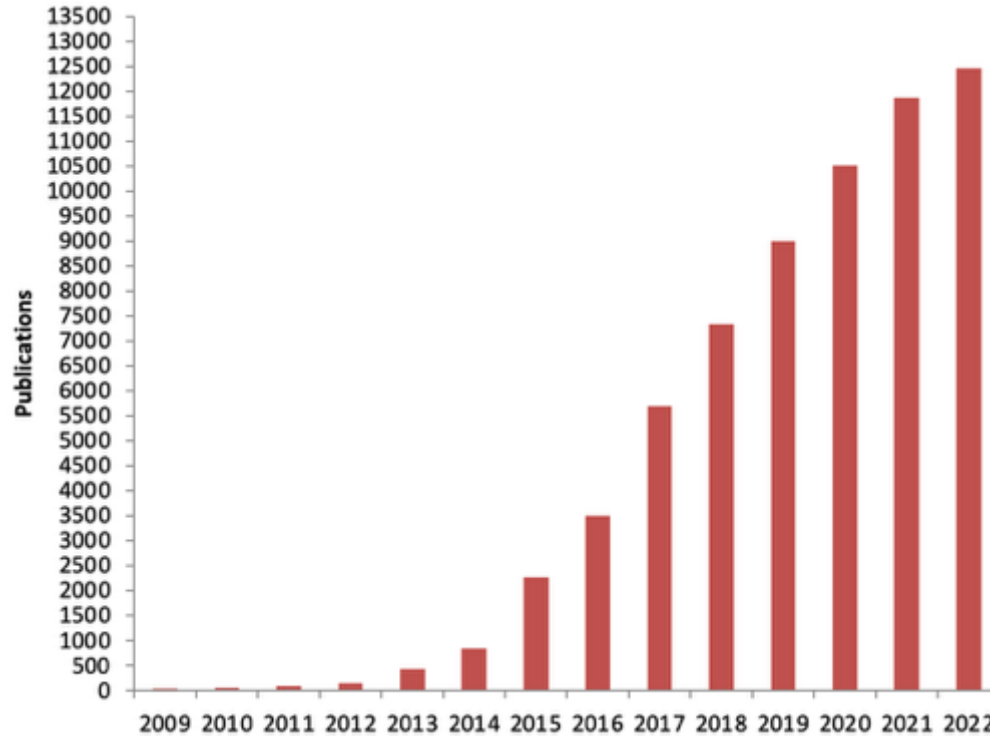
Woong Y Hwang^{1,7}, Yanfang Fu^{2,3,7}, Deepak Reyon^{2,3}, Morgan L Maeder^{2,4}, Shengdar Q Tsai^{2,3}, Jeffry D Sander^{2,3}, Randall T Peterson^{1,5,6}, J-R Joanna Yeh^{1,5} & J Keith Joung²⁻⁴

Jan 29th 2013



Science
RNA-Guided
Engineering

Prashant Mali,^{1,5} L.
James E. DiCarlo,¹



Engineering Using

Lin,^{1,5} Robert Barretto,⁶
ran Jiang,⁶ Luciano A.

system

35 papers per day!

Nearly 1% of all PubMed citations

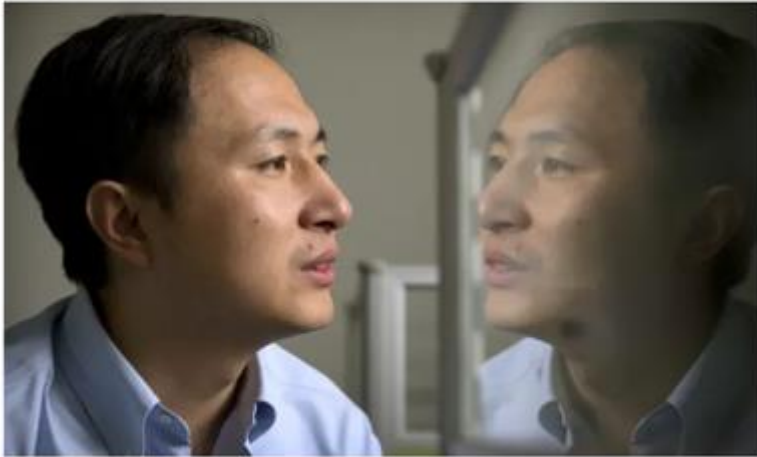
Wenqiang Wang,^{1,2,3} Shengdar Q. Tsai,^{2,3},
Morgan L. Maeder^{2,4}, Shengdar Q. Tsai^{2,3},
Jeffrey D. Sander^{2,3}, Randall T. Peterson^{1,5,6},
J-R Joanna Yeh^{1,5} & J. Keith Joung²⁻⁴

Germline therapy

theguardian

'Of course it's not ethical': shock at gene-edited baby claims

Chinese geneticist He Jiankui's claim to have altered embryos prompts outcry from scientists



▲ He Jiankui. Chinese authorities have ordered an investigation to verify his claims. Photograph: Mark Schiefelbein/AP

nature
International journal of science

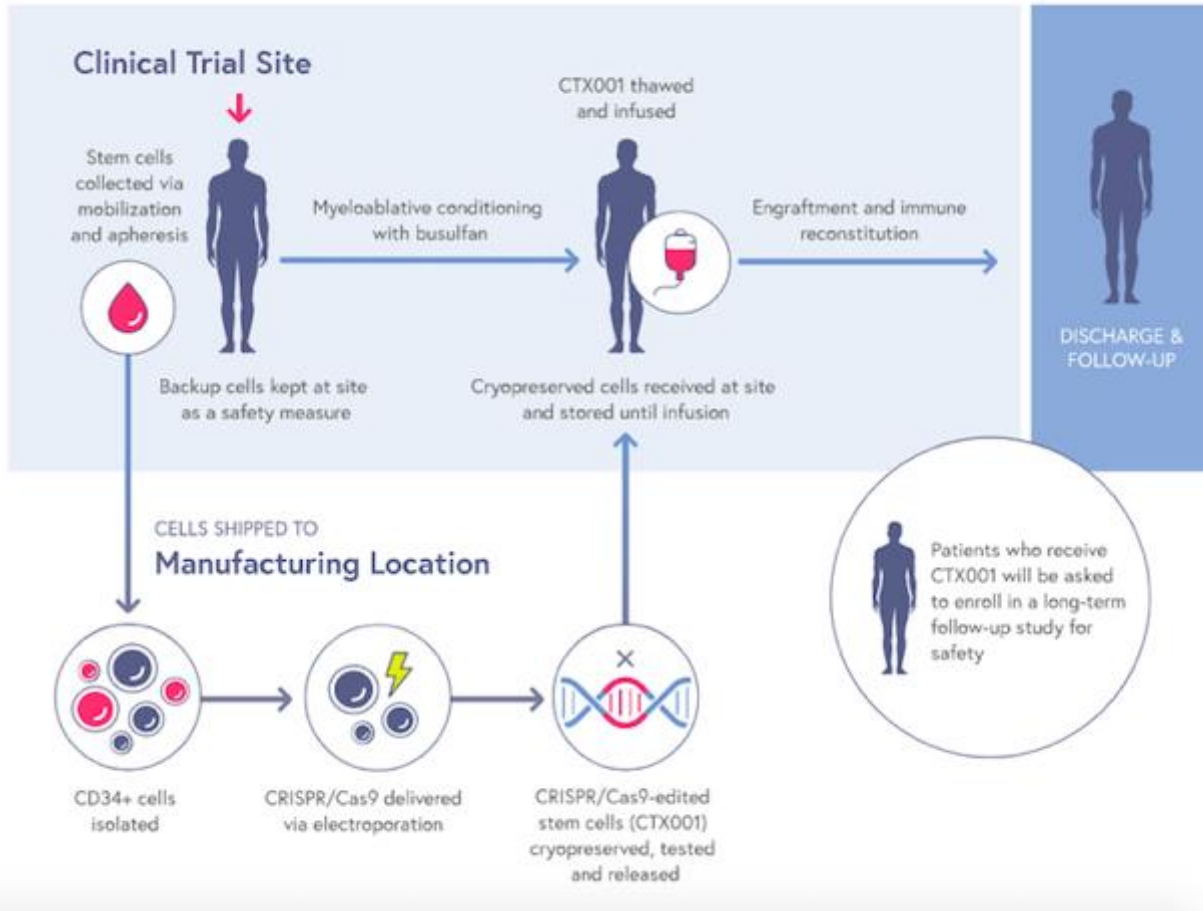
NEWS • 30 NOVEMBER 2018

First CRISPR babies: six questions that remain



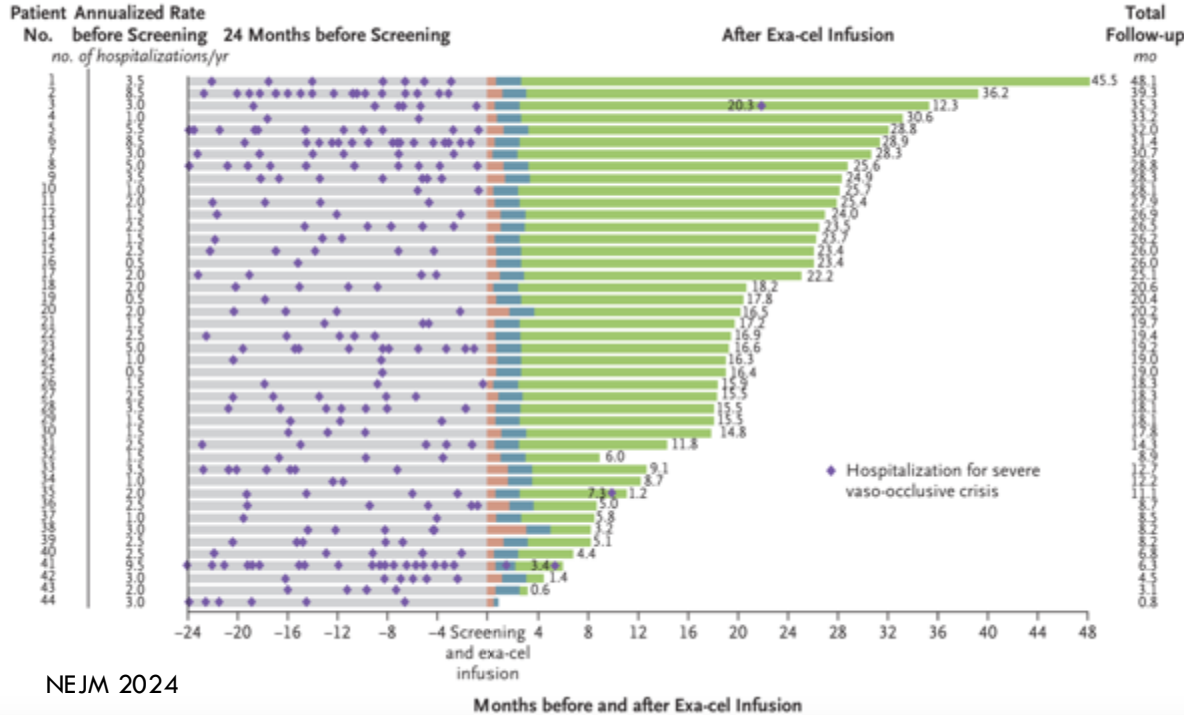
2018 – Edited HIV receptor to prevent HIV infection of babies

Sickle Cell Anaemia



CASGEVY - Sickle Cell Anaemia

B Duration of Periods Free from Inpatient Hospitalization for Severe Vaso-Occlusive Crises in All Patients



NEJM 2024

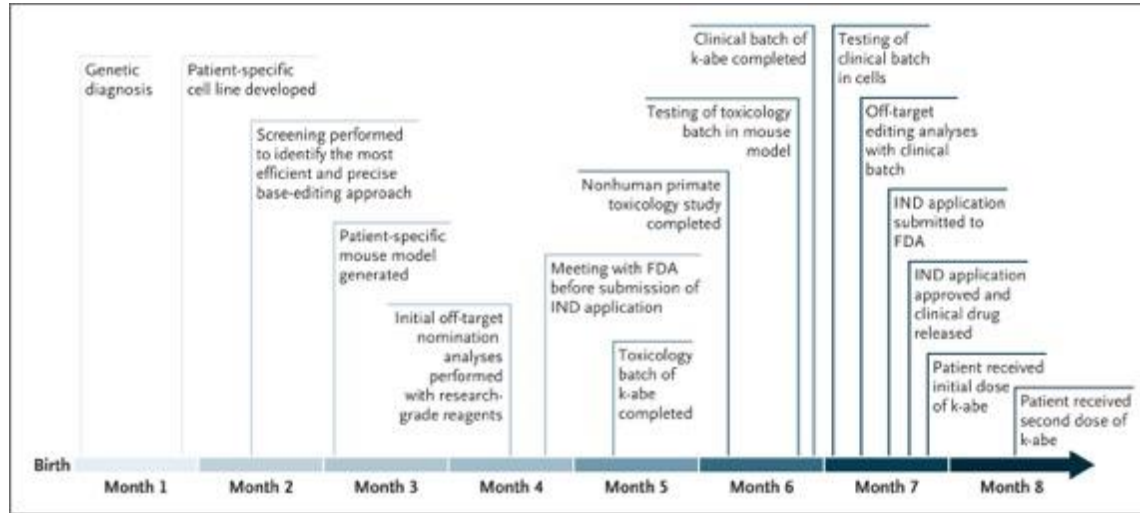
2015: Cell line editing shows upregulation of HbF

2020: Proof of principle in 2x patients

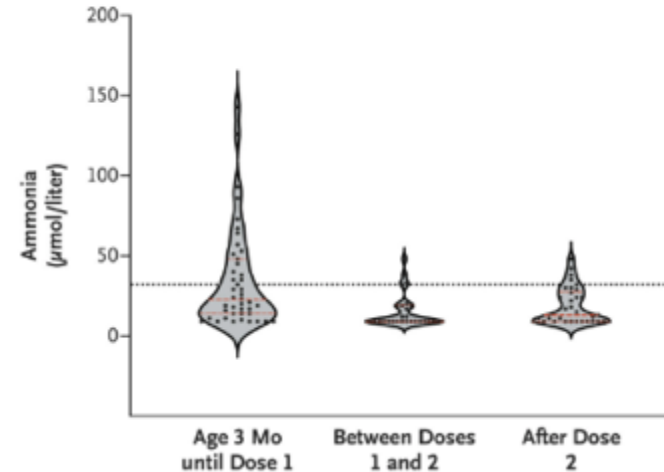
2022: After treatment 30/31 (97%) sickle cell patients had no VOC

2023: CASGEVY approved by UK MHRA

K-abe – Rare disease treatment in 8 months!



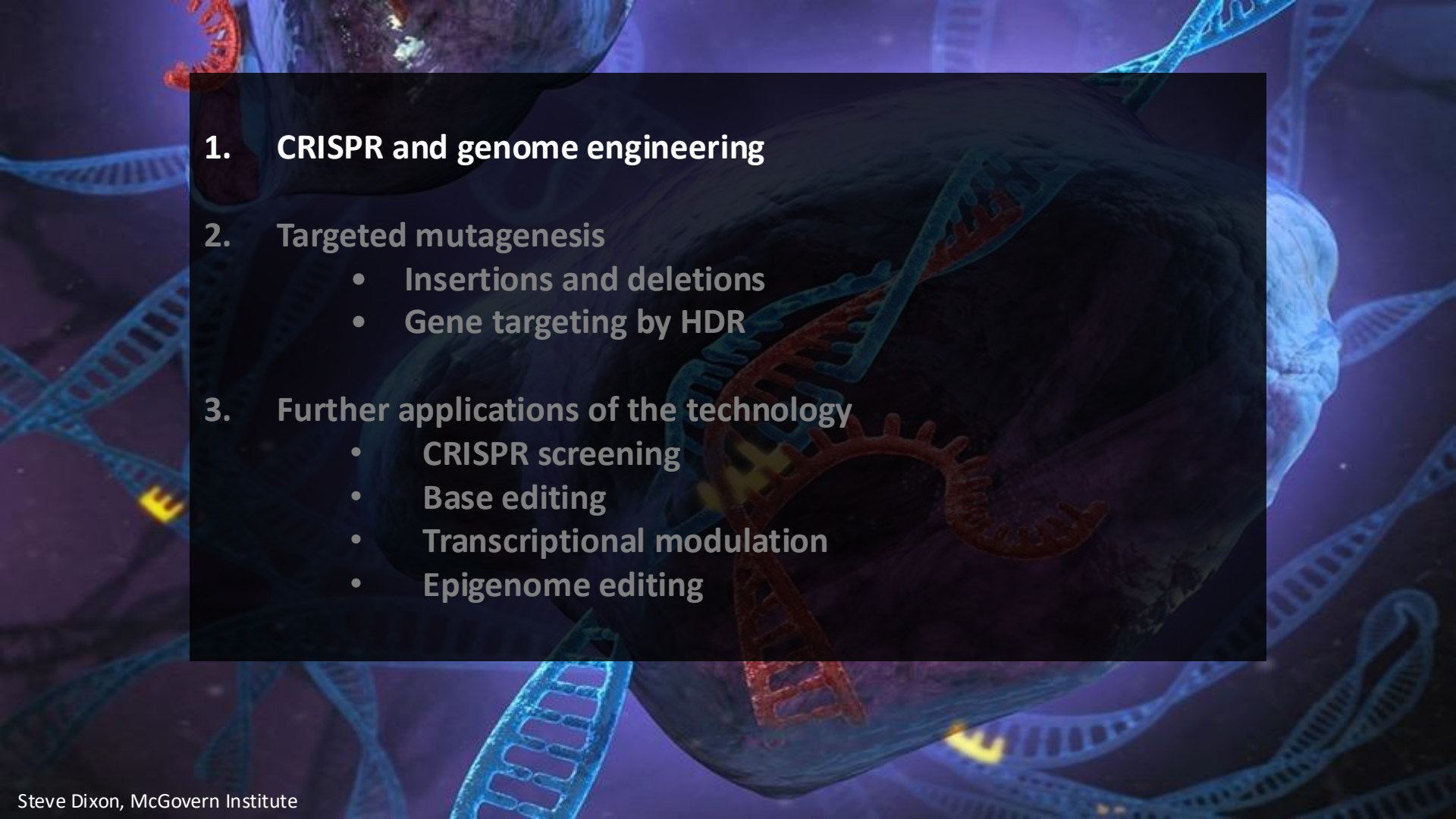
E Plasma Ammonia Levels before and after Treatment

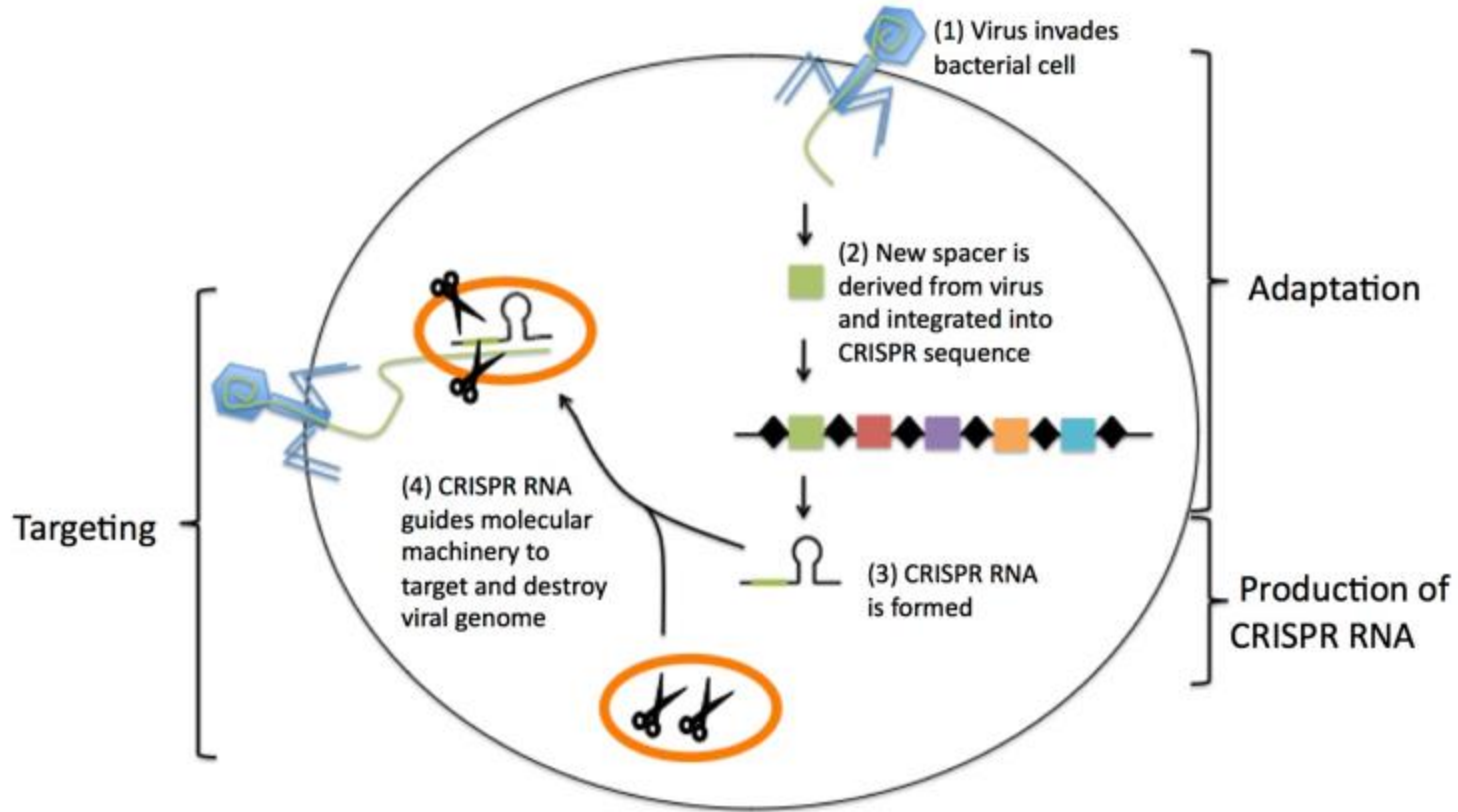


Carbamoyl-phosphate synthetase 1 (CPS1) deficiency (stop codon)

Require liver transplant, but ~50% patients die before they are able to do so

Adenine base editor delivered as mRNA using LNP

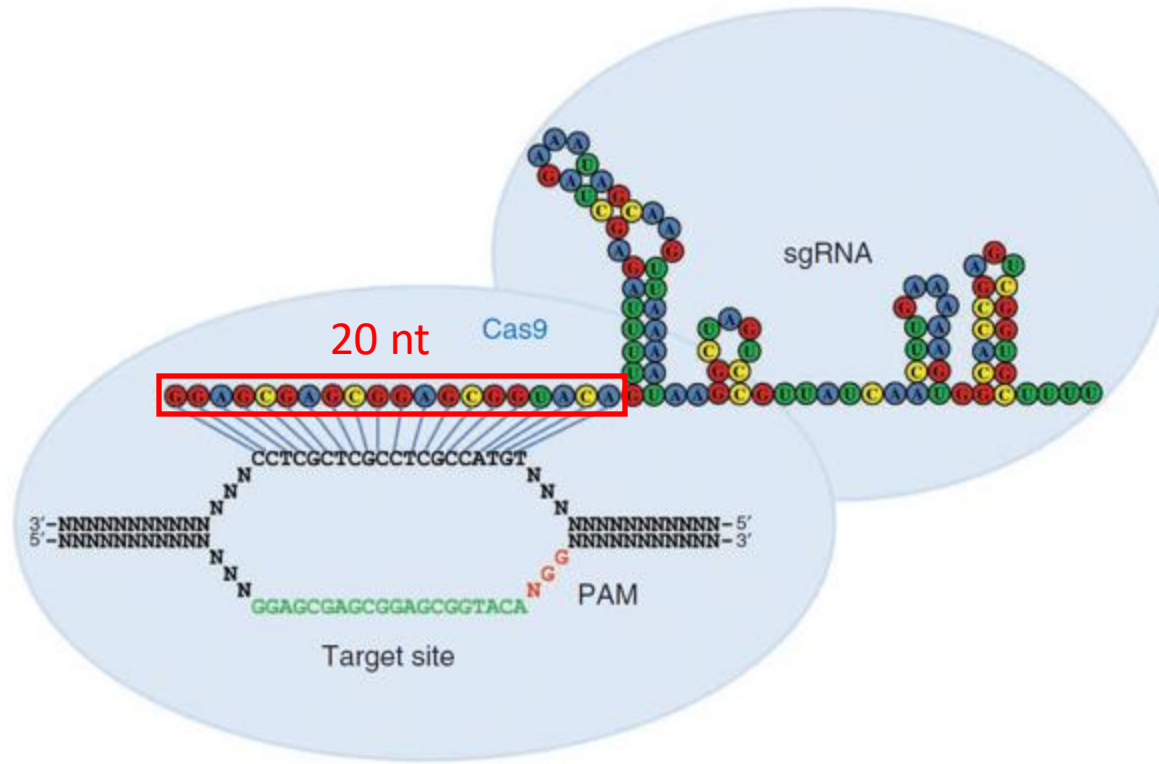
- 
- 1. CRISPR and genome engineering**
 - 2. Targeted mutagenesis**
 - Insertions and deletions
 - Gene targeting by HDR
 - 3. Further applications of the technology**
 - CRISPR screening
 - Base editing
 - Transcriptional modulation
 - Epigenome editing



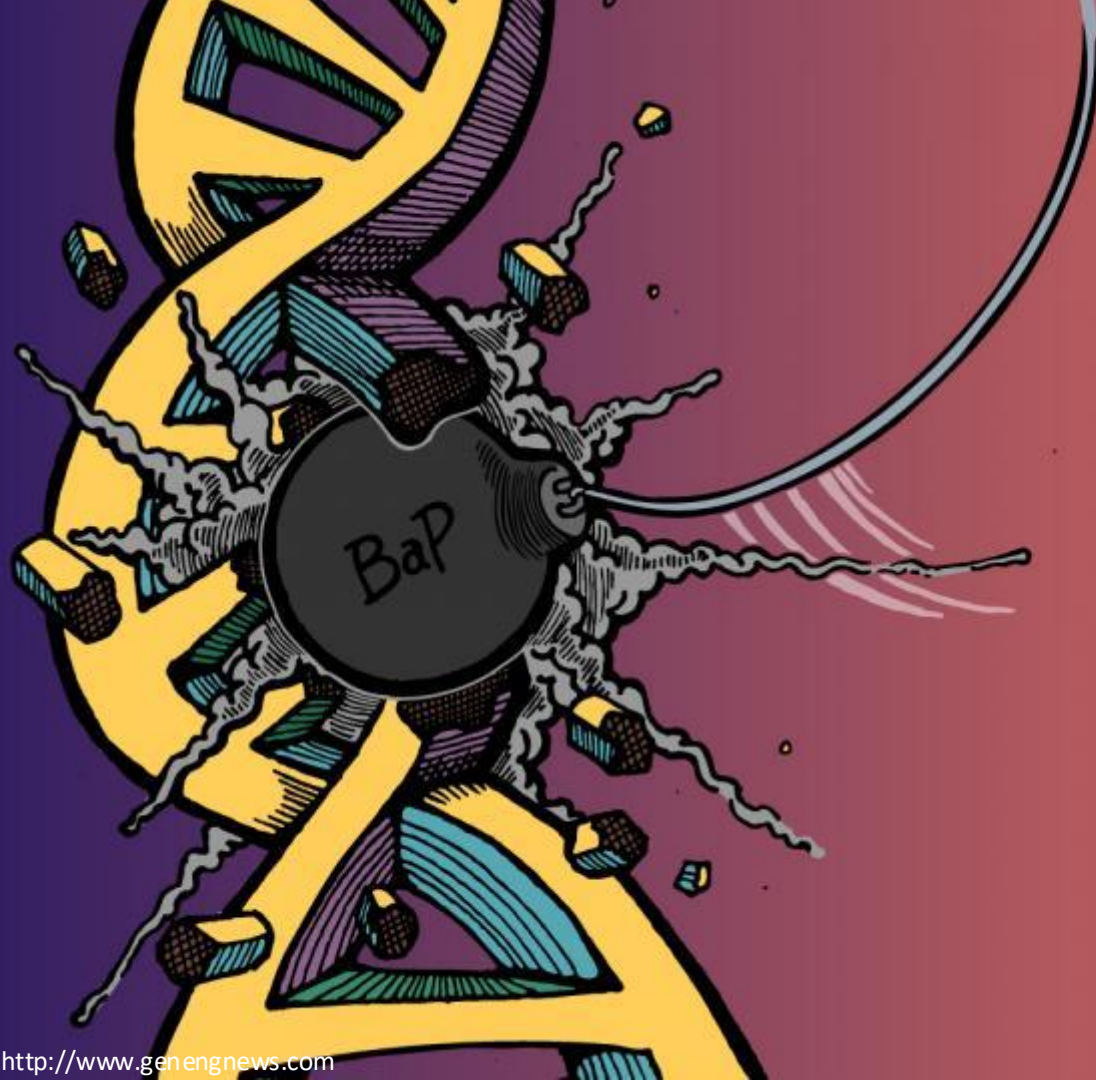
<http://sitn.hms.harvard.edu>

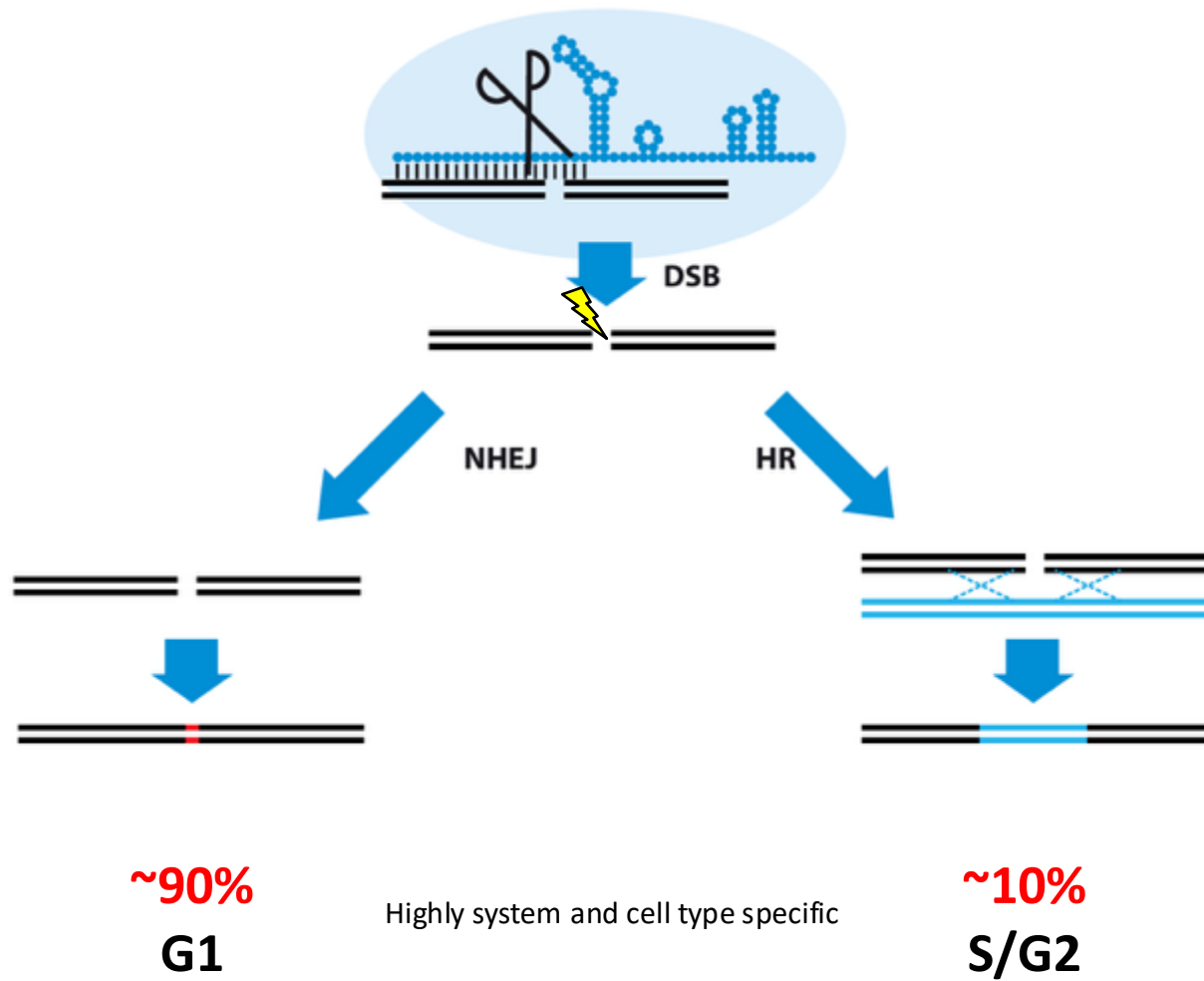
CRISPR = clustered, regularly interspaced, short palindromic repeats

RNA-guided site-specific DNA cleavage Gasiunas et al. PNAS 2012, Jinek et al. Science 2012

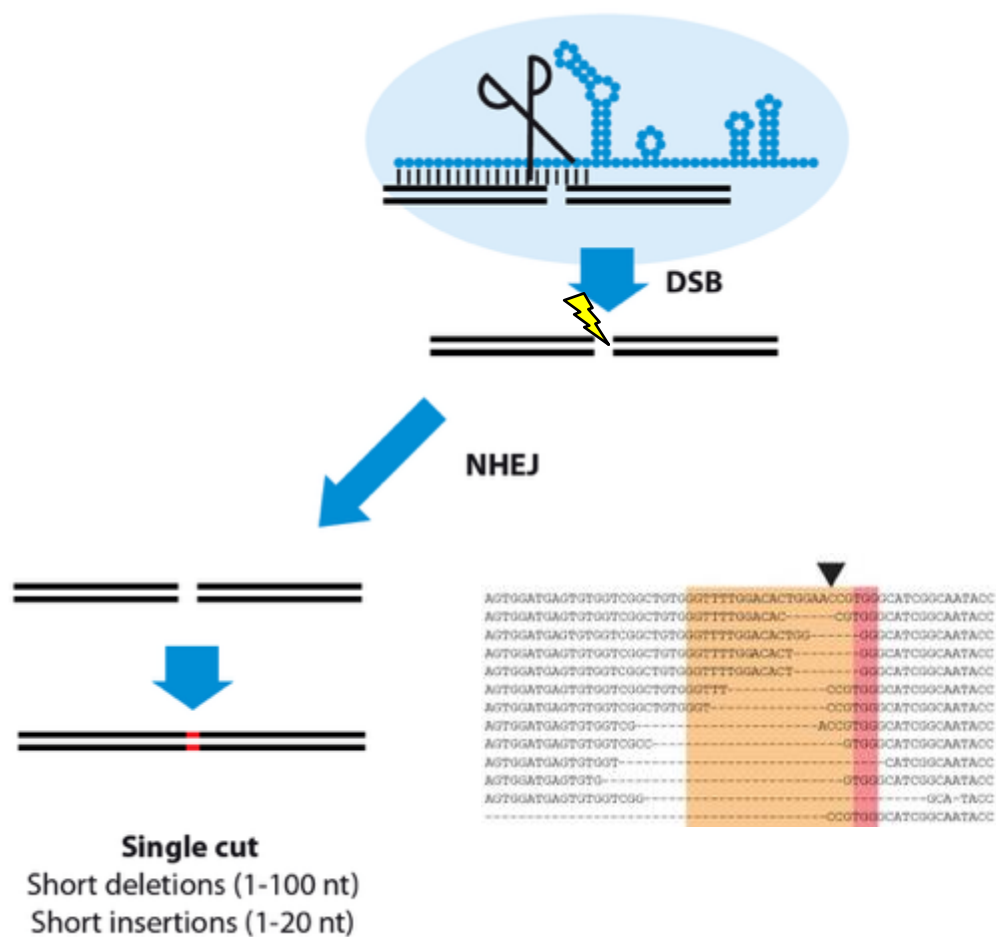




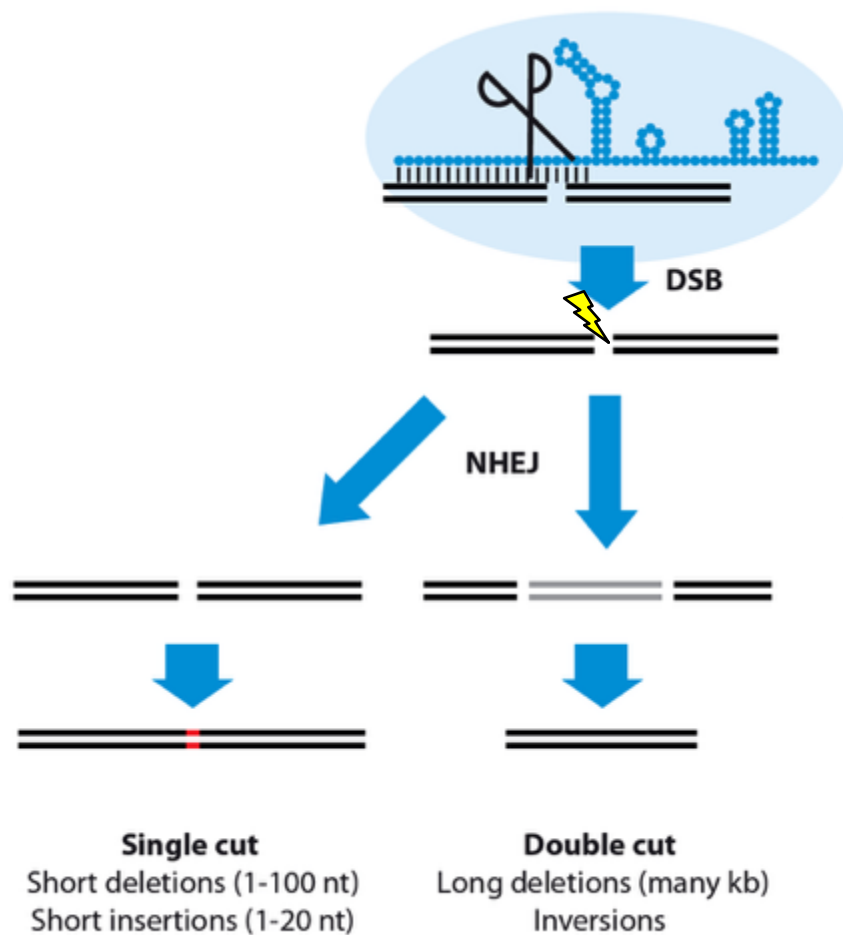


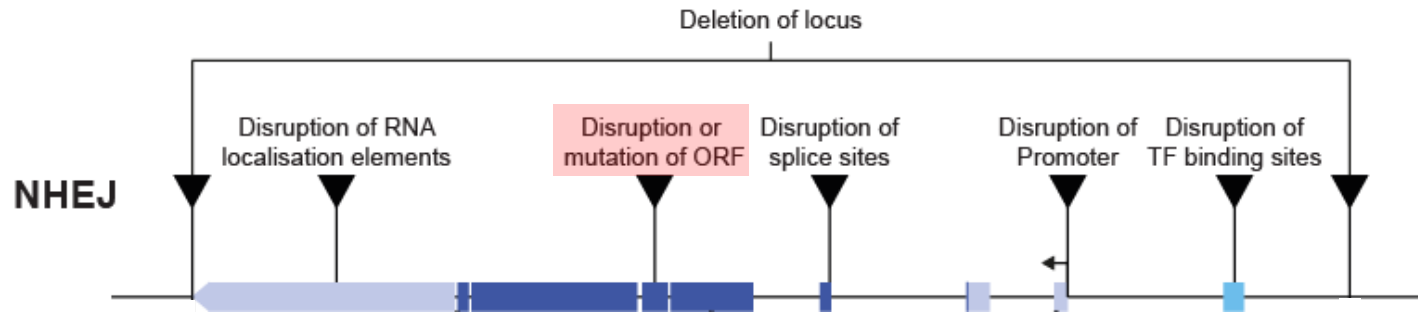


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 - **Other Cas variants**
 - **Base and prime editing**
 - **Transcriptional modulation**
 - **Epigenome editing**

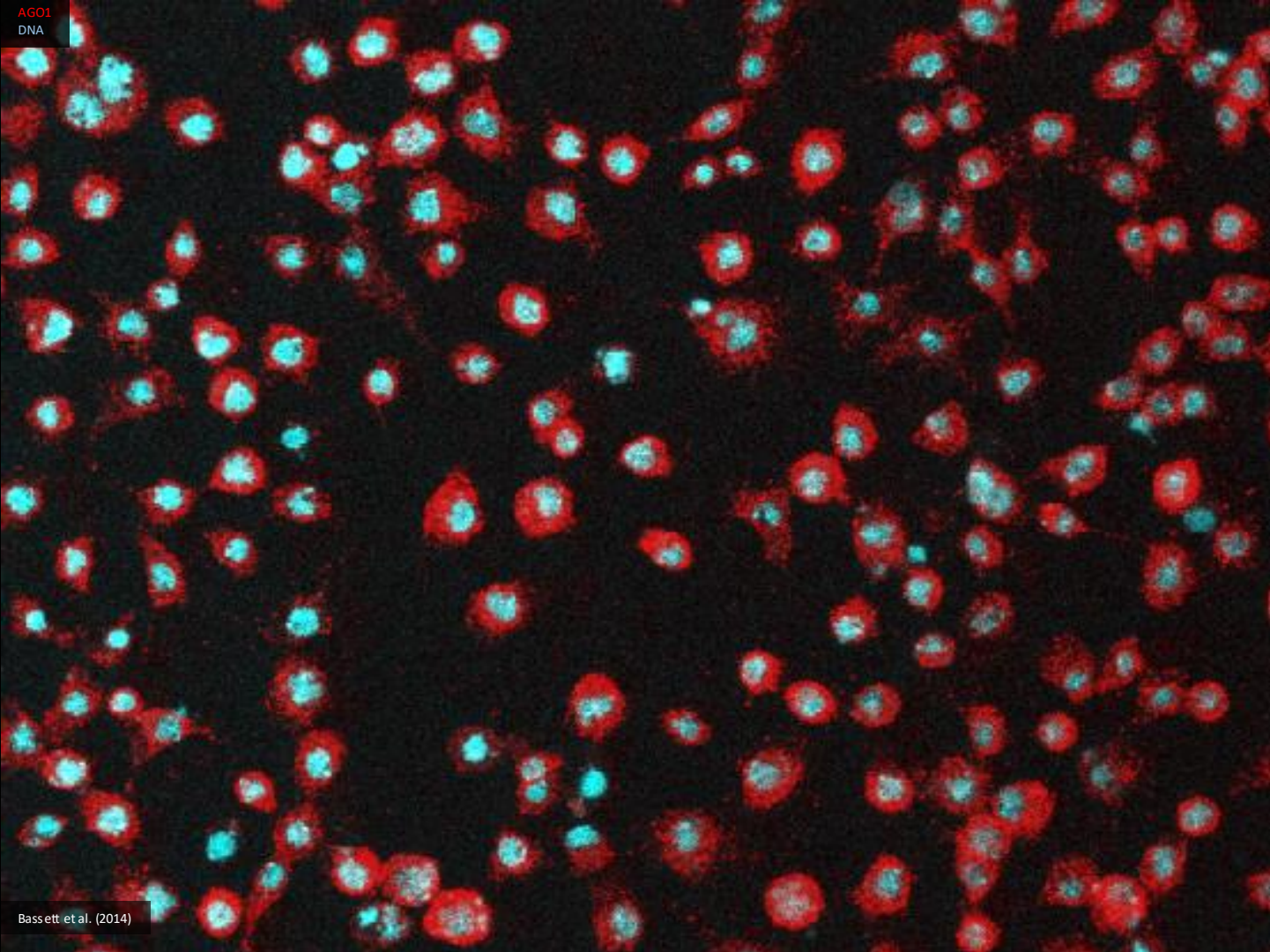


Single cut
Short deletions (1-100 nt)
Short insertions (1-20 nt)



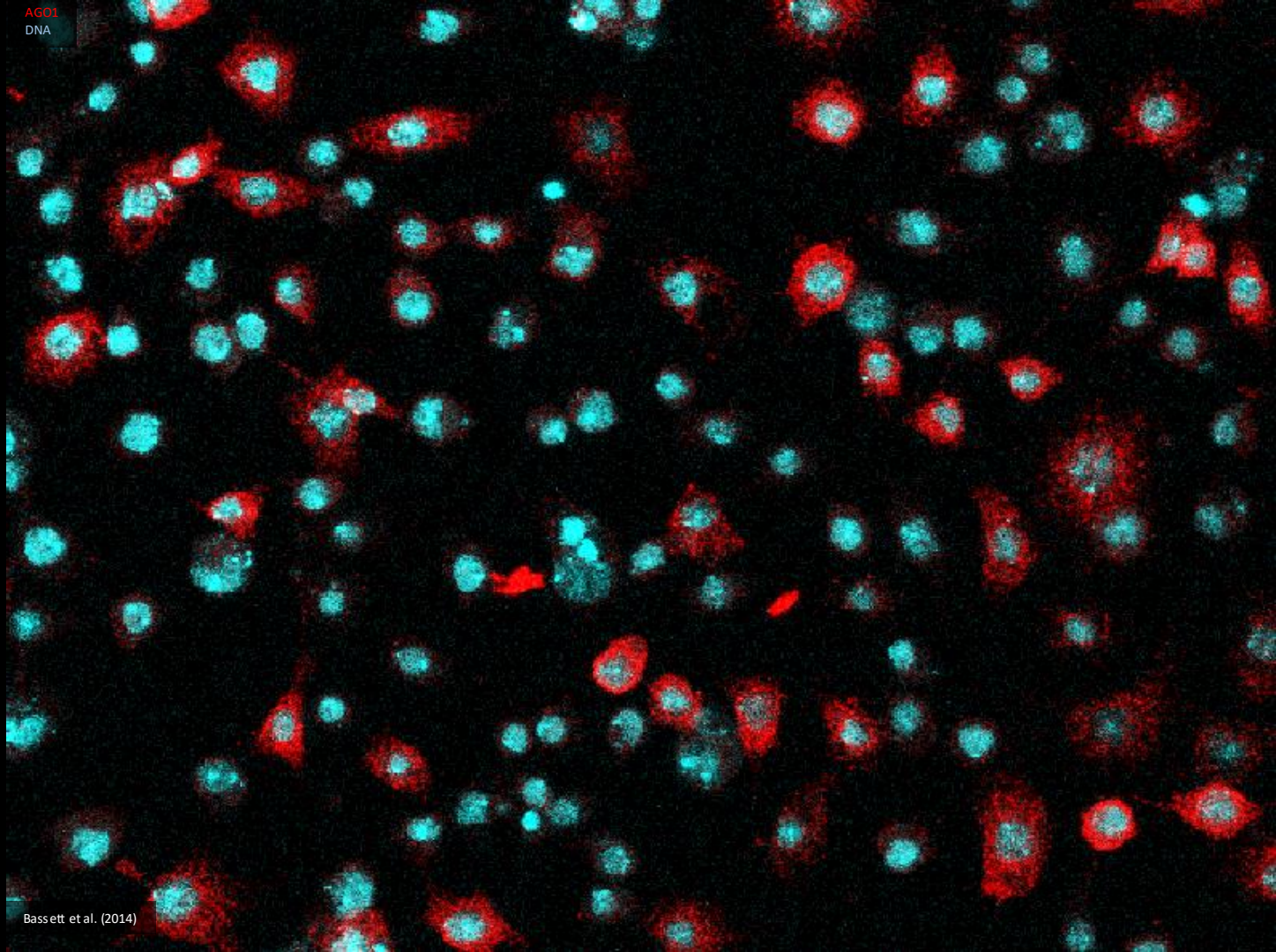


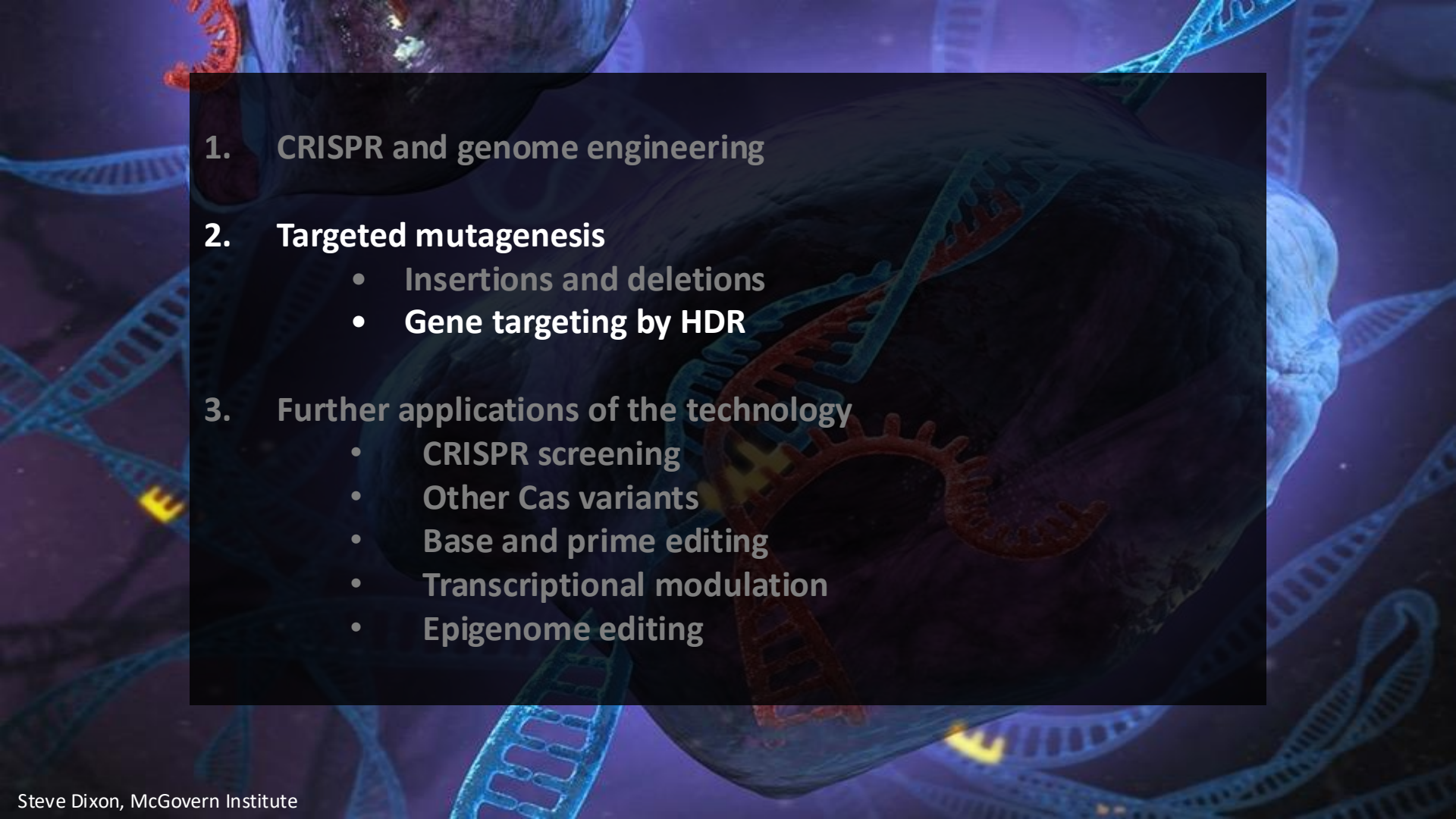
AGO1
DNA

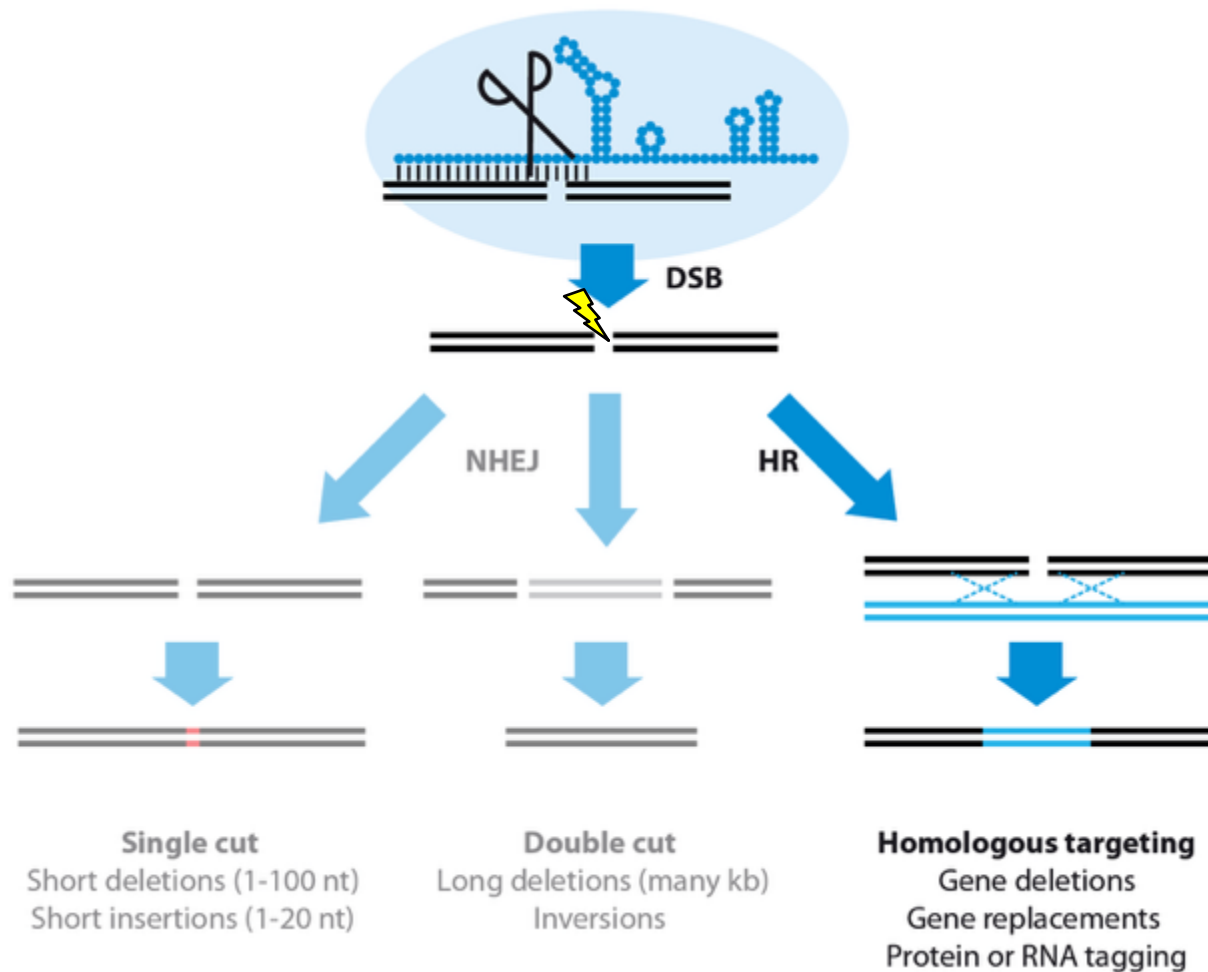


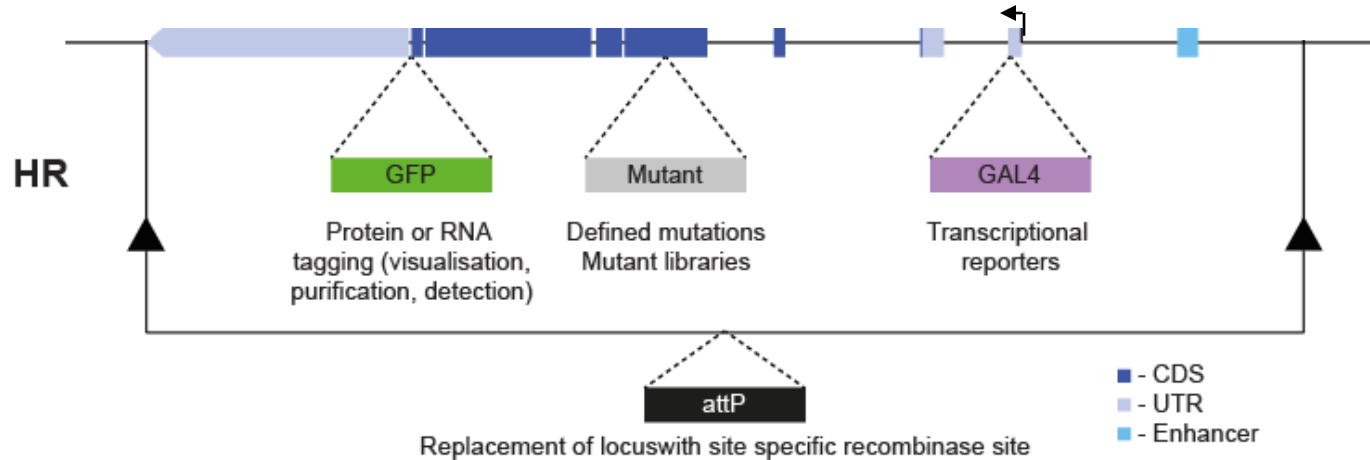
Bassett et al. (2014)

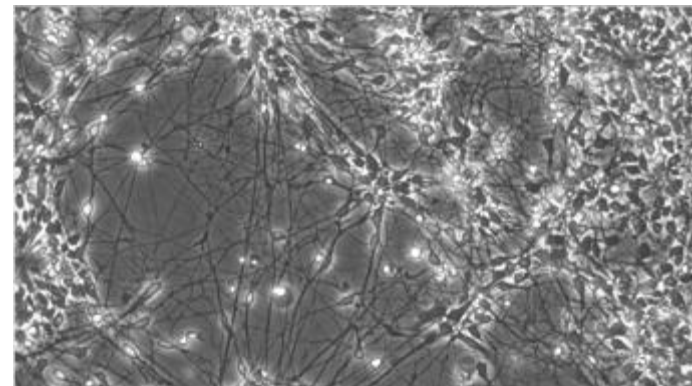
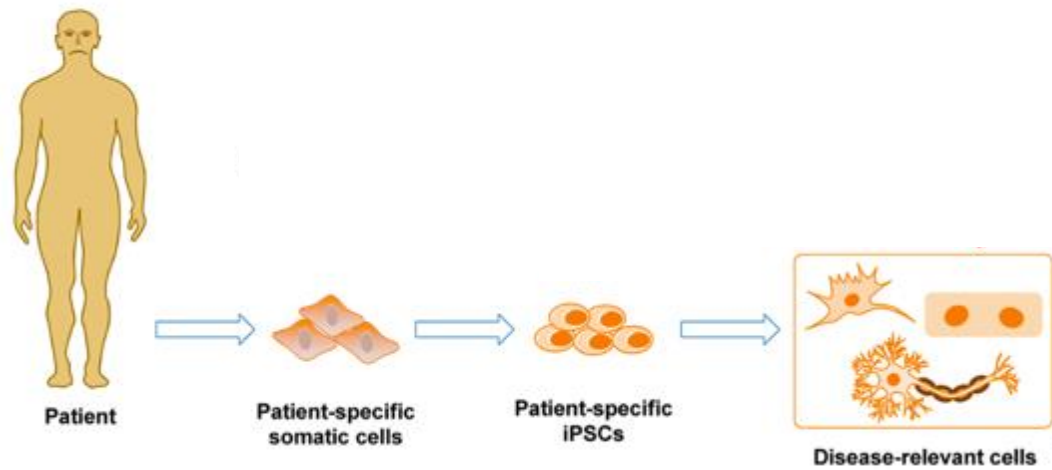
AGO1
DNA

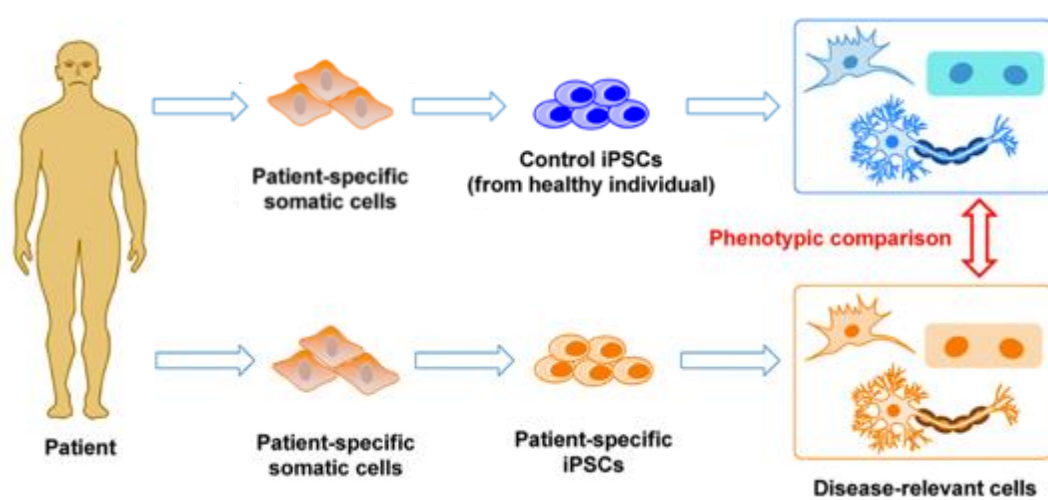


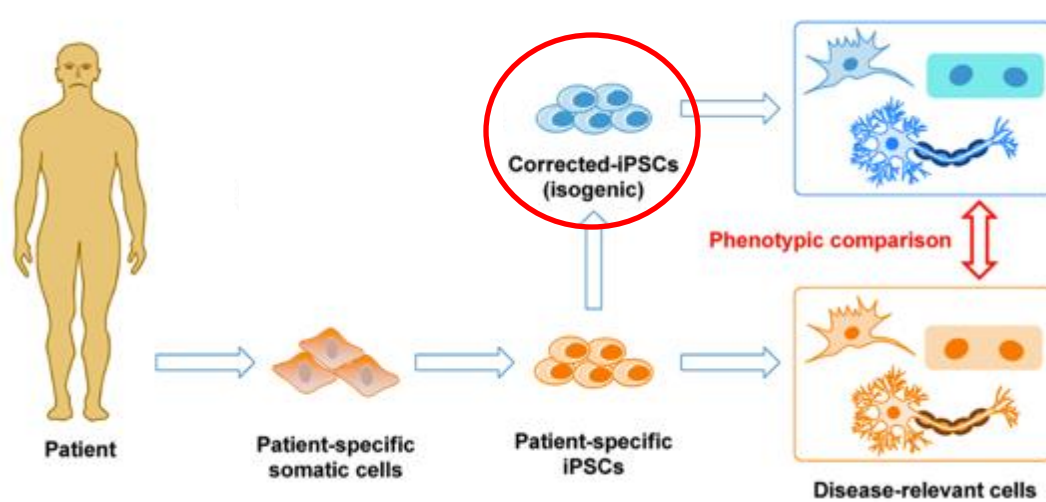
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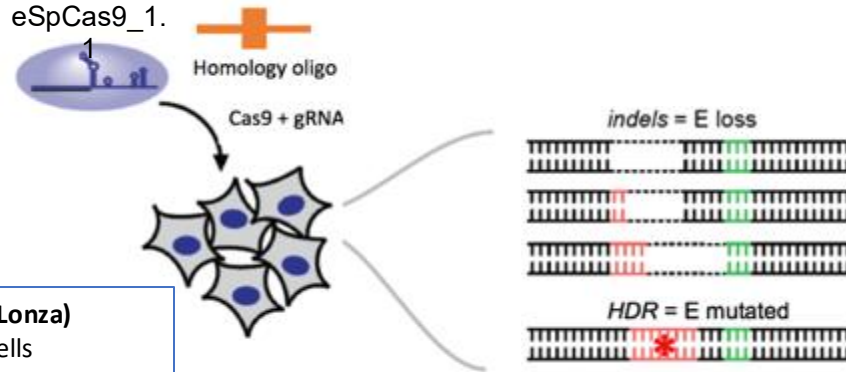






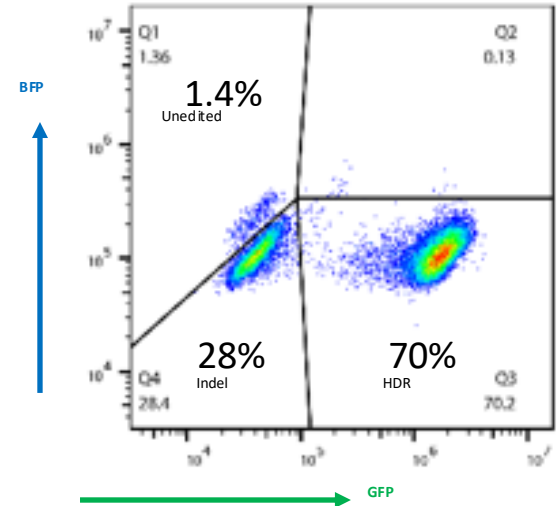
- Eliminate genetic background variability between “WT” control lines
- Enables more subtle changes to be seen
- Establish true causative lesion

Isogenic hiPSCs



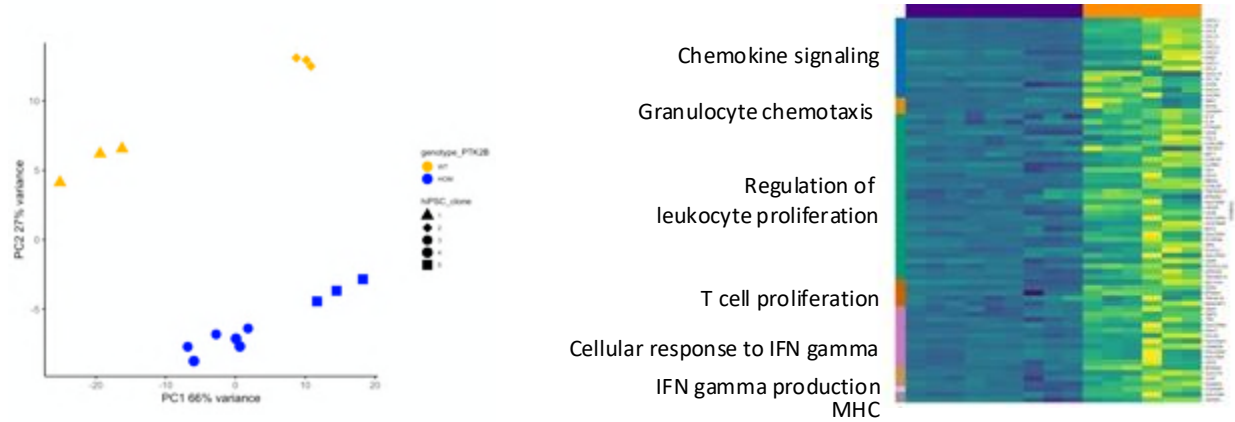
Amara 4D (Lonza)

1 x 10⁶ cells
 120 pmol Cas9 protein
 250 pmol Synthetic sgRNA
 500 pmol 100 nt ssDNA oligo
 AZD DNAPKi, cold shock

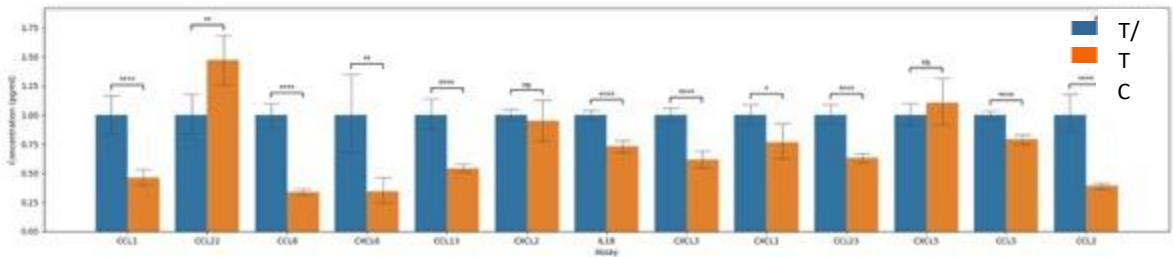


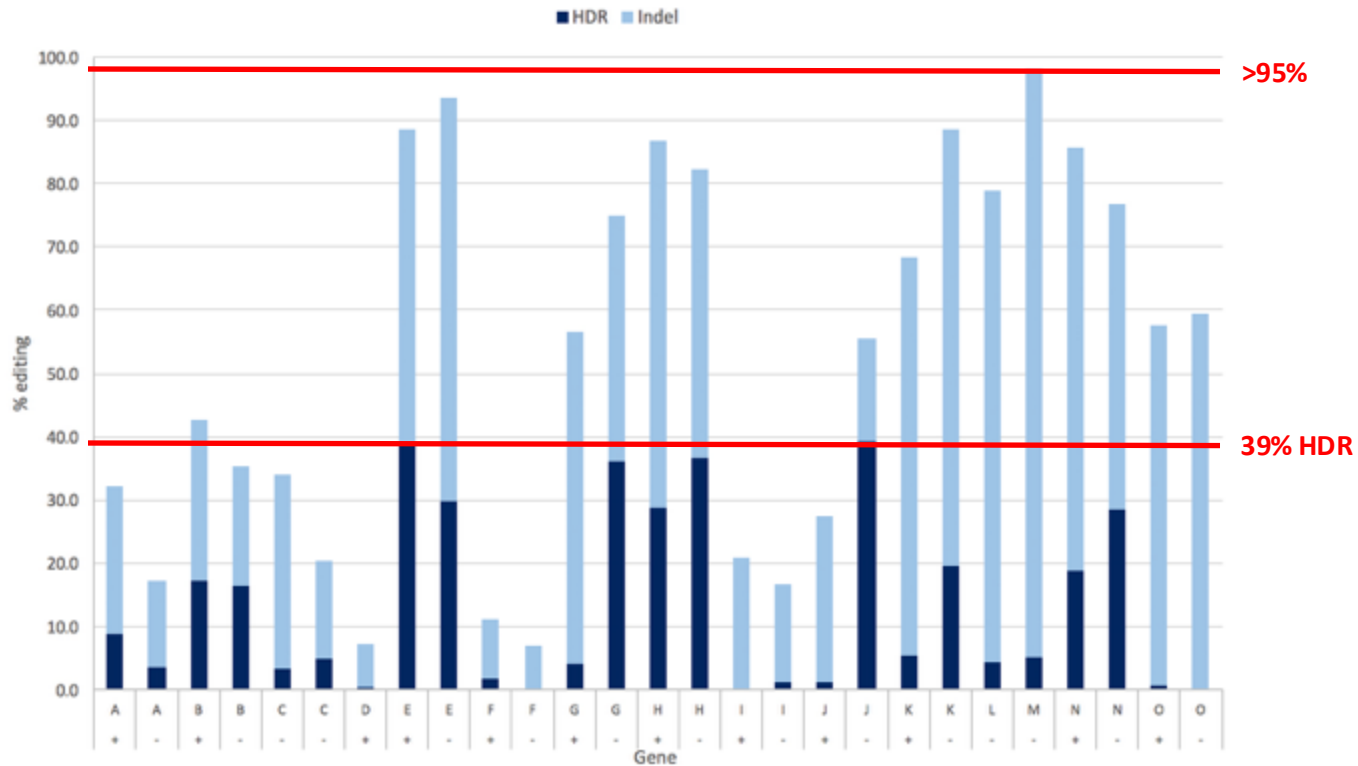
PTK2B intronic variant

RNAseq

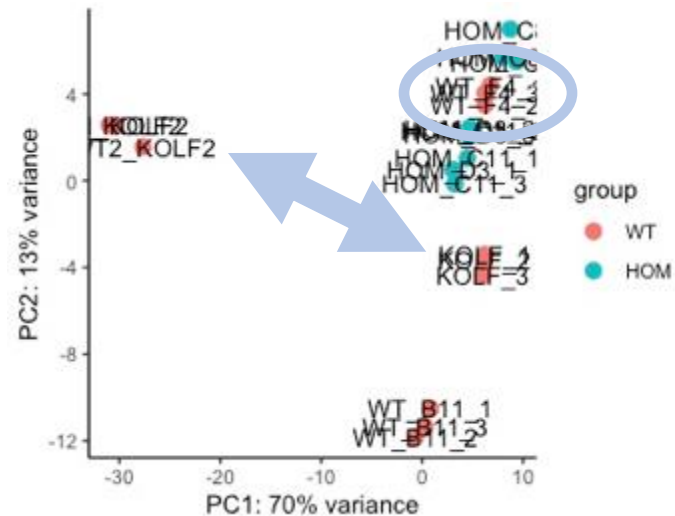
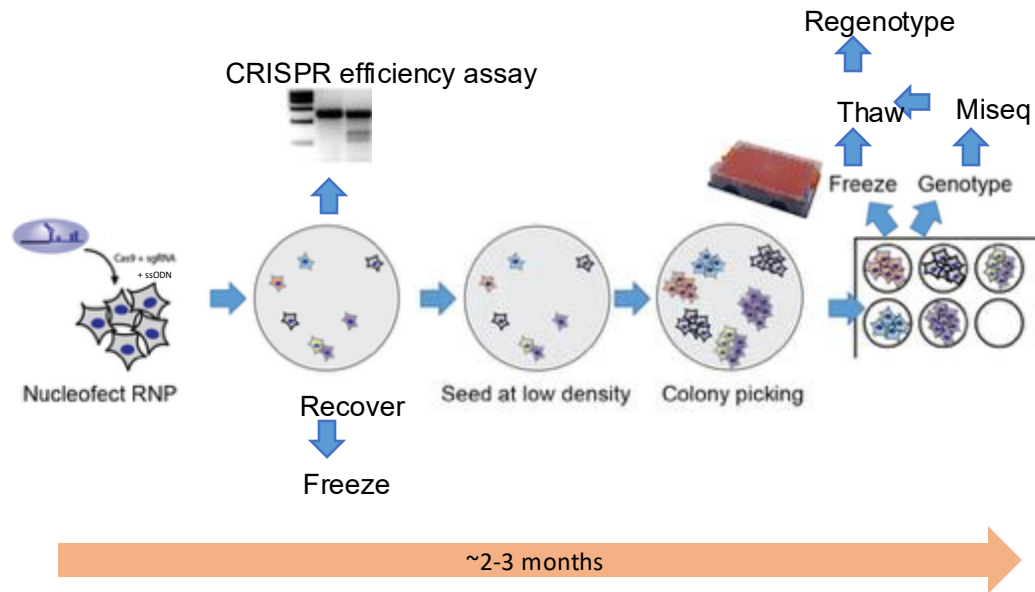


Chemokine release





- Efficiency down to <1% in worst cases

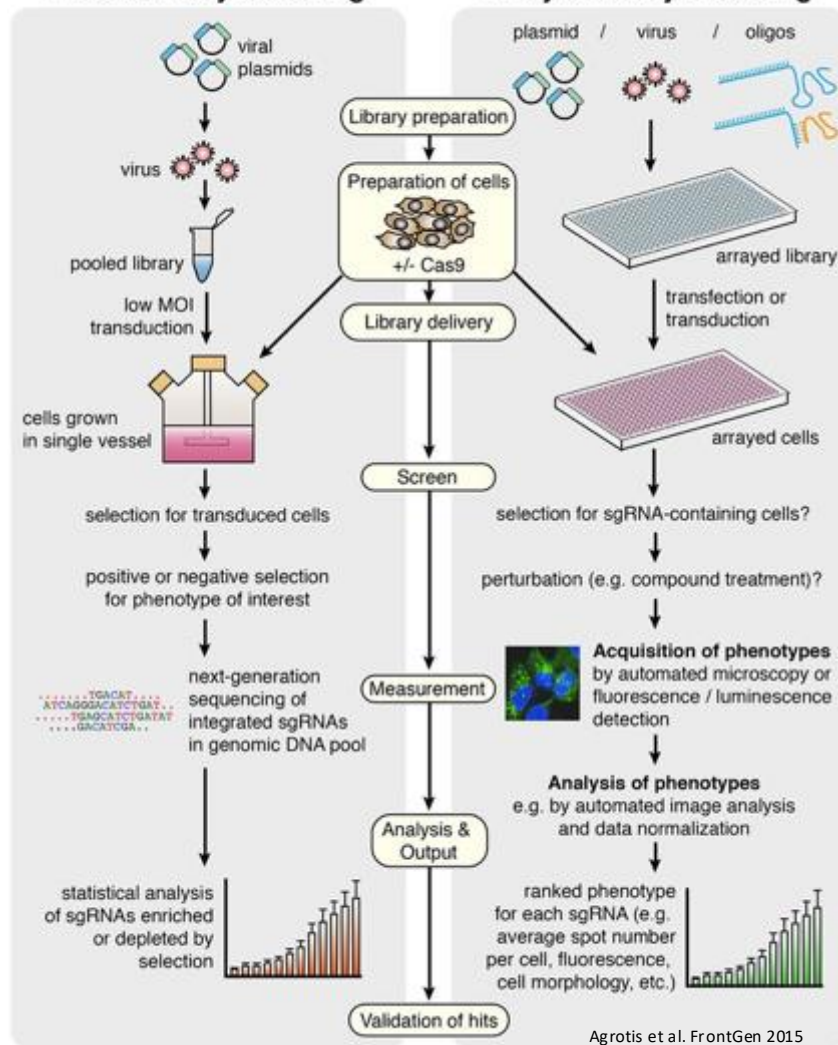


- Time and cost
- Multiple clones
- Variability between clones
- Variability in differentiation
- ... Not feasible for screening

- 
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Pooled library screening

Arrayed library screening

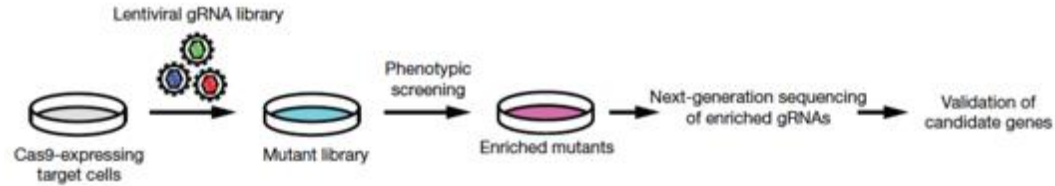


- Does not require robotics
- Limited to single cell assays (e.g. life/death, FACS, scRNAseq, OPS)

- Complex and multidimensional readouts e.g. high content imaging
- Limited number of genes, or robotics required for larger screens

Genome-wide recessive genetic screening in mammalian cells with a lentiviral CRISPR-guide RNA library

Hiroko Koike-Yusa^{1,2}, Yilong Li^{1,2}, E-Pien Tan¹, Martin Del Castillo Velasco-Herrera¹ & Kosuke Yusa¹



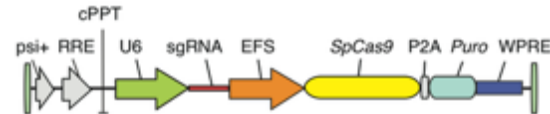
Genetic Screens in Human Cells Using the CRISPR/Cas9 System

Tim Wang,^{1,2,3,4} Jenny J. Wei,^{1,2} David M. Sabatini,^{1,2,3,4,5*}† Eric S. Lander^{1,3,6*}†



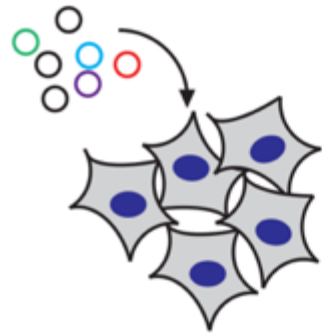
Genome-Scale CRISPR-Cas9 Knockout Screening in Human Cells

Ophir Shalem,^{1,2*} Neville F. Sanjana,^{1,2*} Ella Hartenian,¹ Xi Shi,^{1,3} David A. Scott,^{1,2} Tarjei Mikkelsen,^{1,2*} Dirk Heckl,⁴ Benjamin L. Ebert,⁴ David E. Root,¹ John G. Doench,² Feng Zhang^{1,2*}



... other better libraries available now e.g. MinLib (Sanger), Brie/Brunello (Broad), Weissman CRISPRa/i

<https://www.addgene.org/pooled-library/>

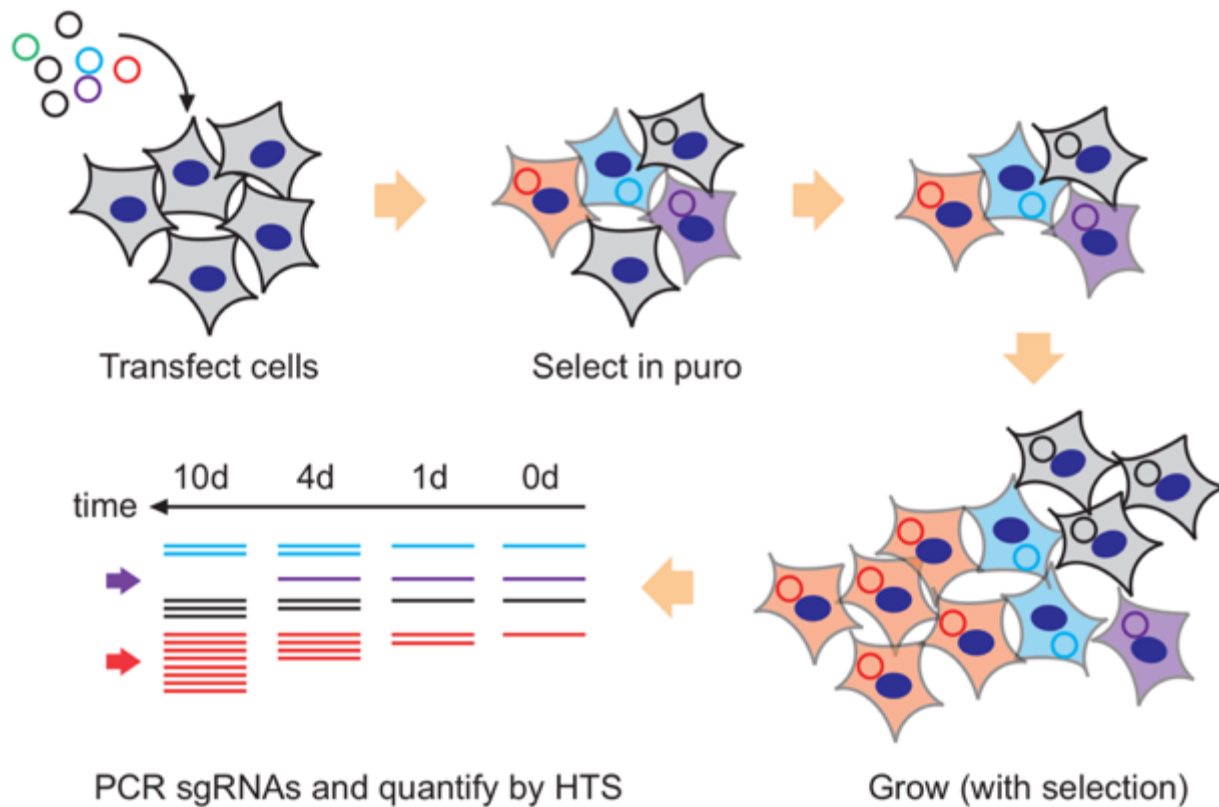


Transfect cells



Select in puro





A large, abstract network diagram in the background, composed of numerous circles of varying sizes (nodes) connected by thin lines (edges). The nodes are colored in shades of pink, purple, and blue, and the overall shape is roughly horizontal, tapering off to the right.

Cancer Dependency Map

Identifying all dependencies in every cancer cell



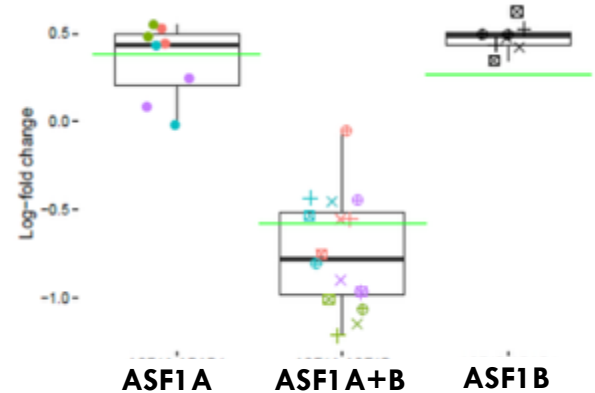
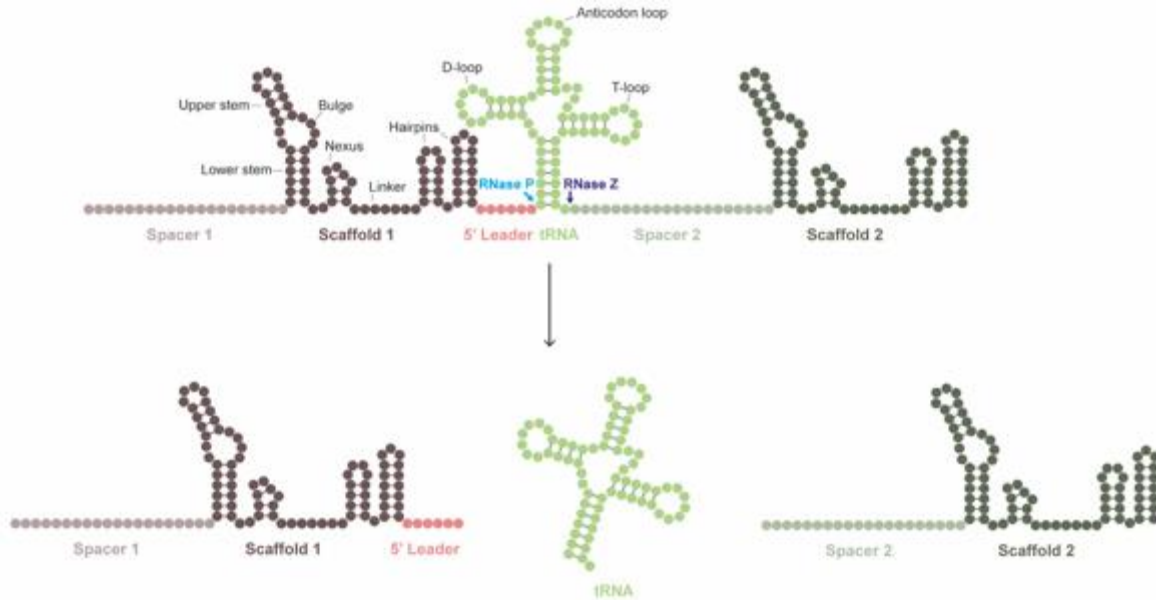
project score

www.depmap.sanger.ac.uk
depmap.org/portal/

789 cancer cell lines screened

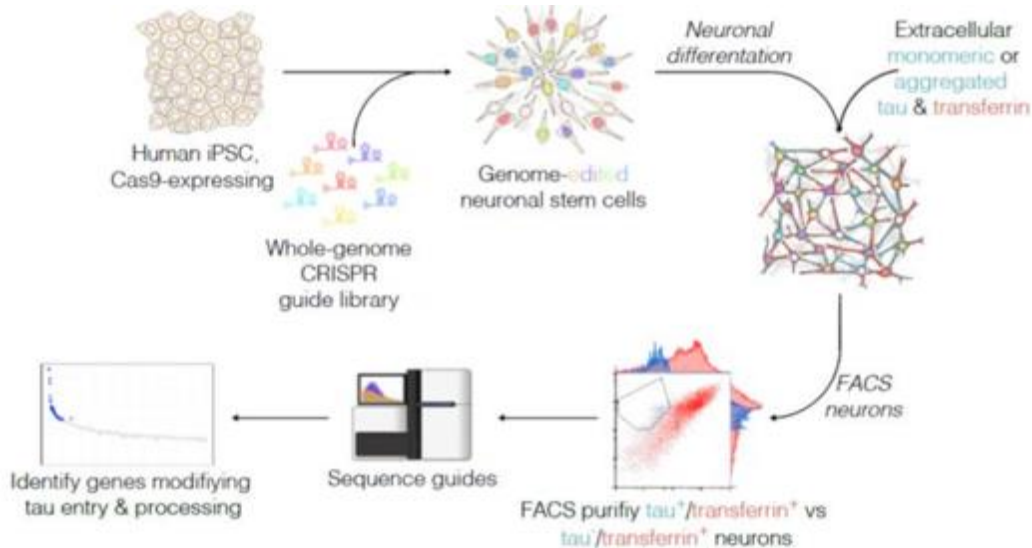
Genetic screens to identify cancer dependencies

Dual perturbation screening



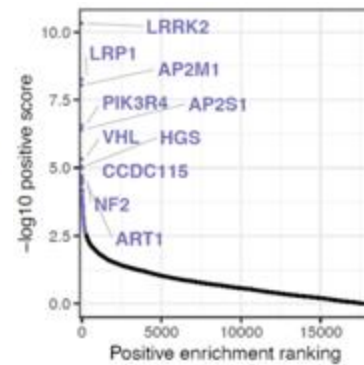
- Synthetic lethal interactions
- Genetic interaction mapping
- Smaller genome-wide libraries

Neuronal uptake of tau

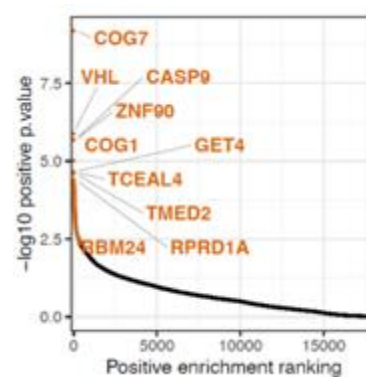


- Top 2 hits: recently discovered tau receptor, the low density lipoprotein receptor LRP1, and PD-associated kinase LRRK2
- Overall similarity with receptor-mediated viral entry screens suggests tau spreading is a quasi-infectious process

Monomeric tau

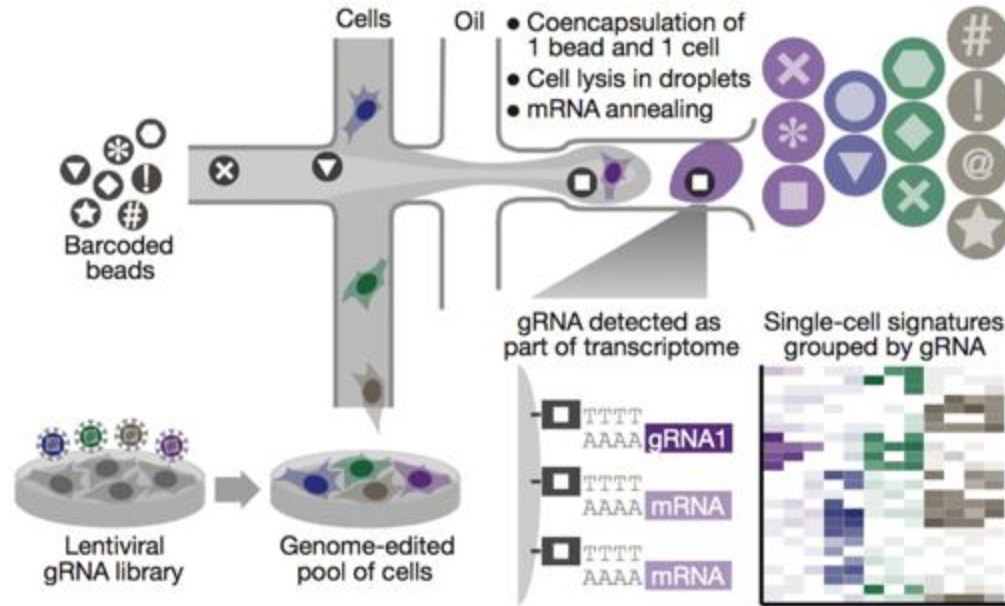


Aggregated tau



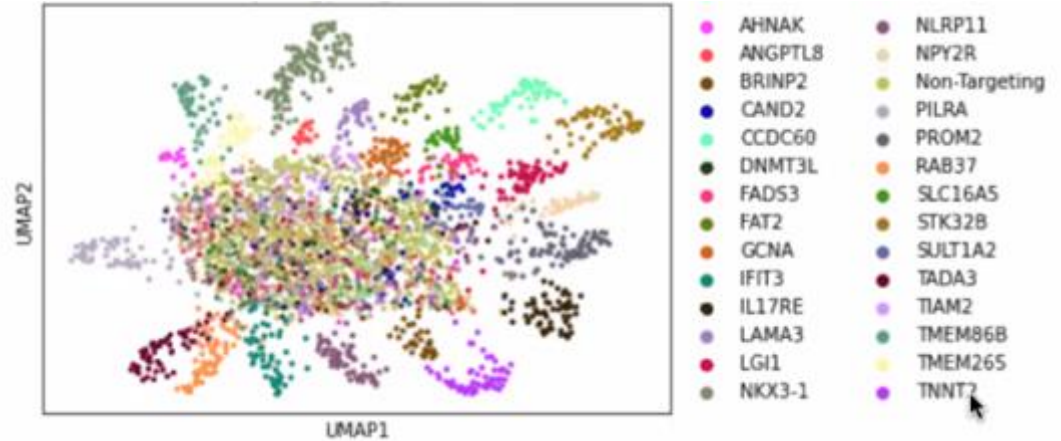
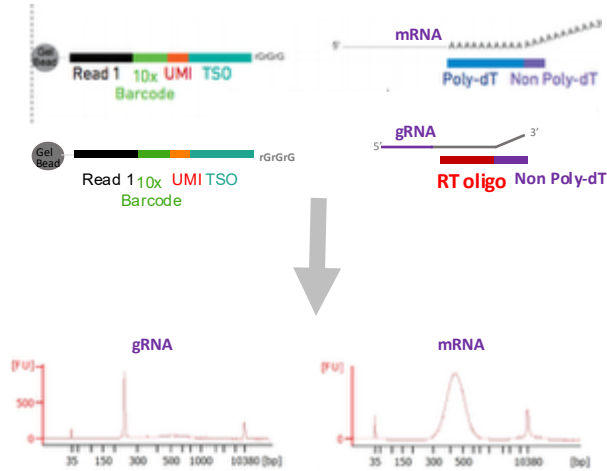
Single cell CRISPR screening

(CROPseq, dc-perturbSeq)

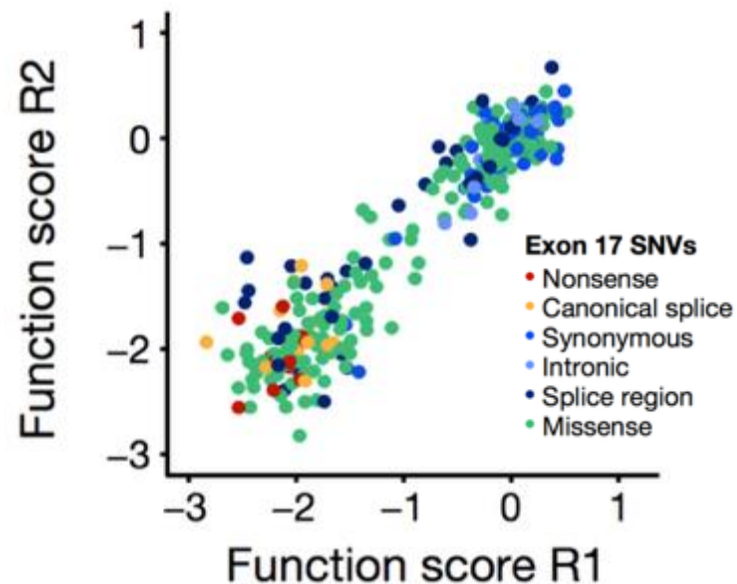
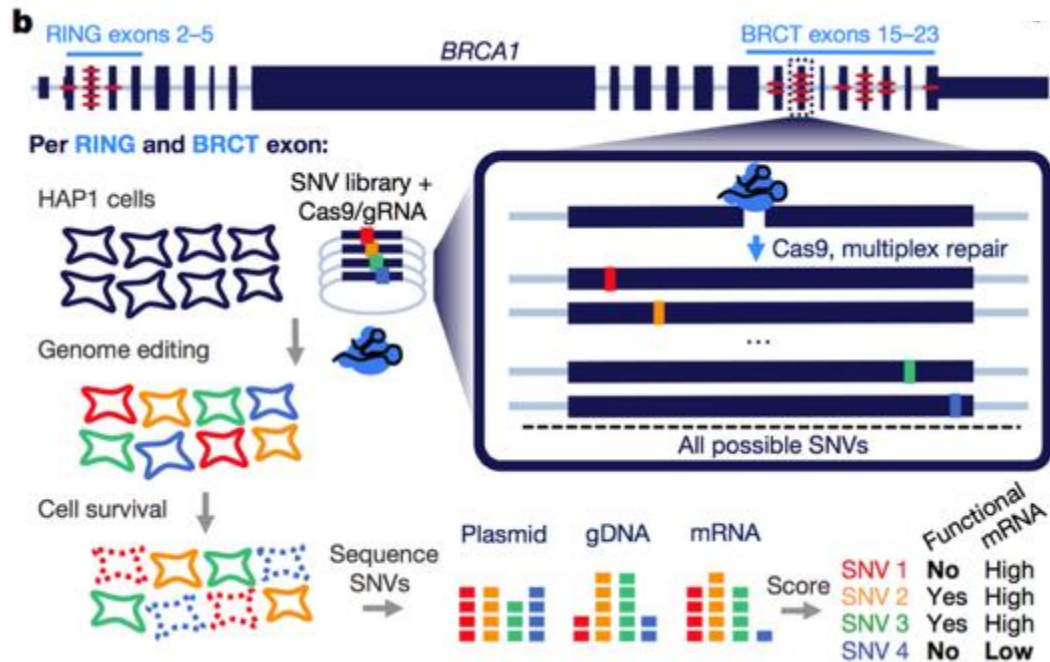


- Complex transcriptomic readout, can be analysed for any gene expression signature
- Can analyse cell-specific effects in mixed populations
- Limited to ~100's of gene knockouts

Single cell CRISPR screening

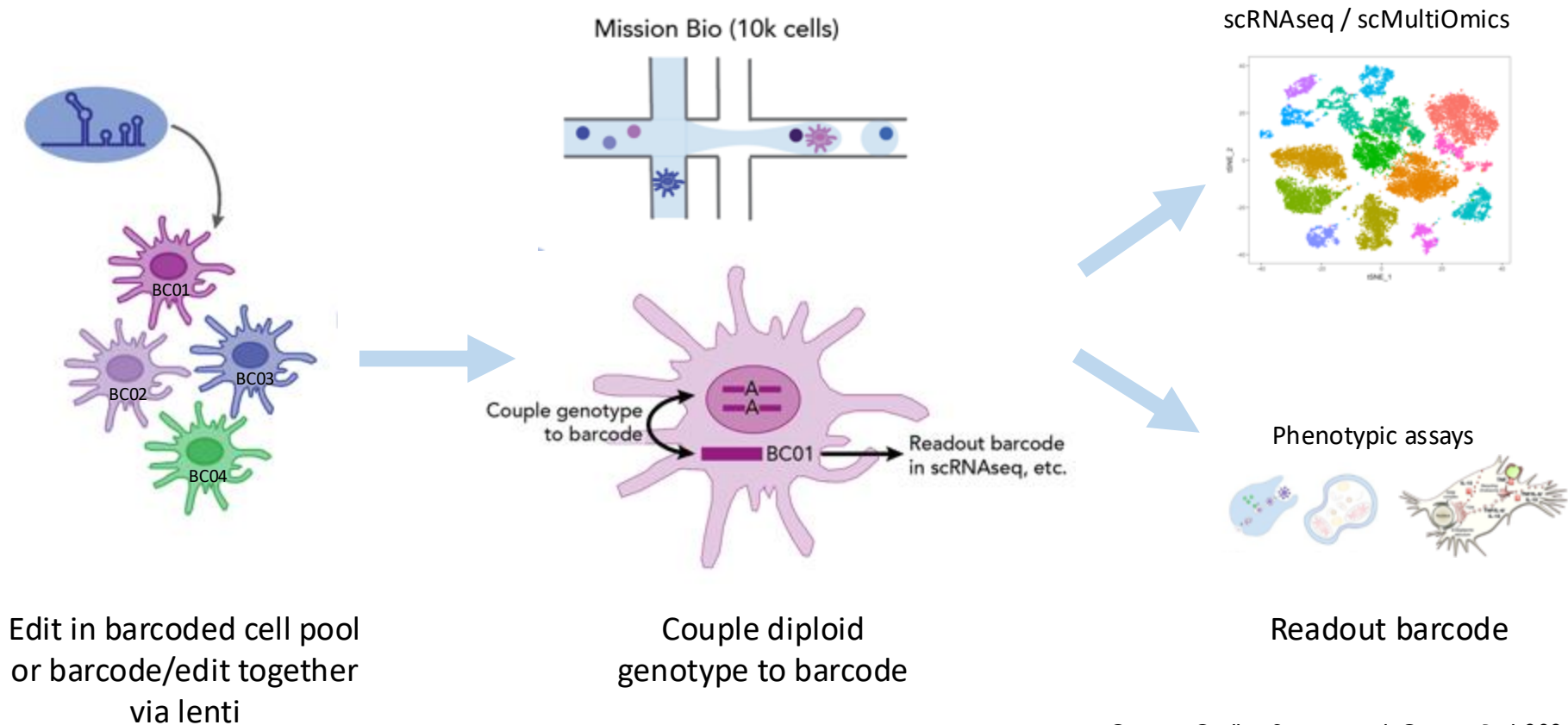


Saturation genome editing

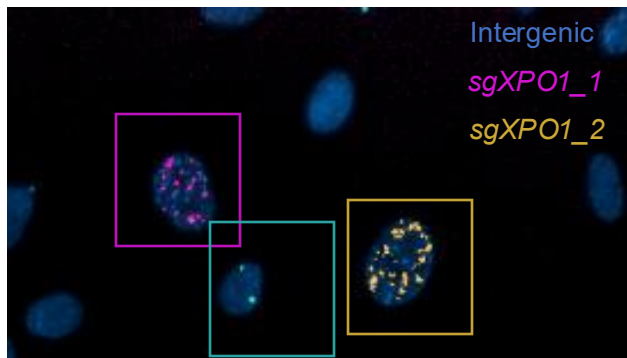


Atlas of Variant Effects

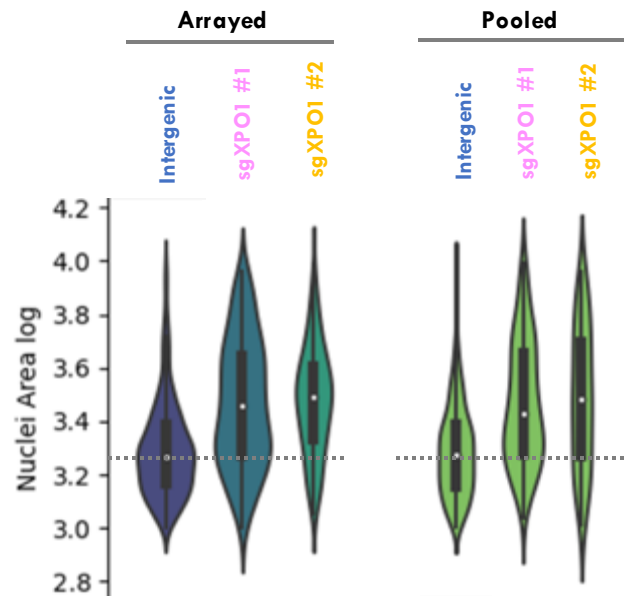
scSNVseq - coupling scGenotype and scRNAseq



Optical Pooled Screening



- KO of XPO1 causes increase in nuclear size



Pooled screening

Perturbations

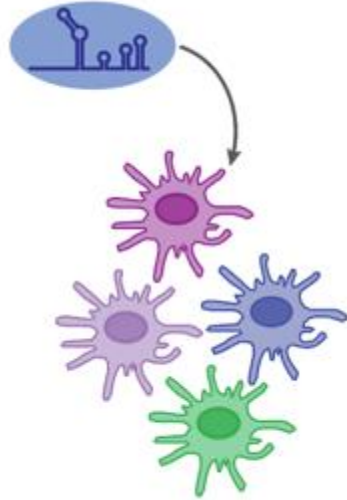
Readouts

RMCE OE

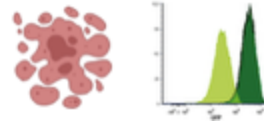
CRISPRa

Dual guide

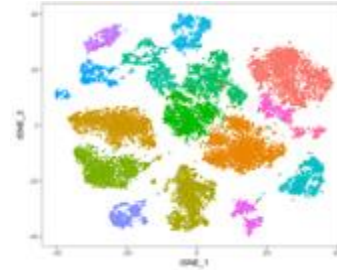
SNPs



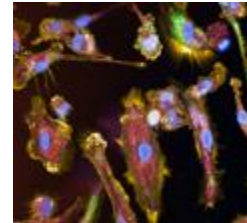
Enrichment phenotypic assays

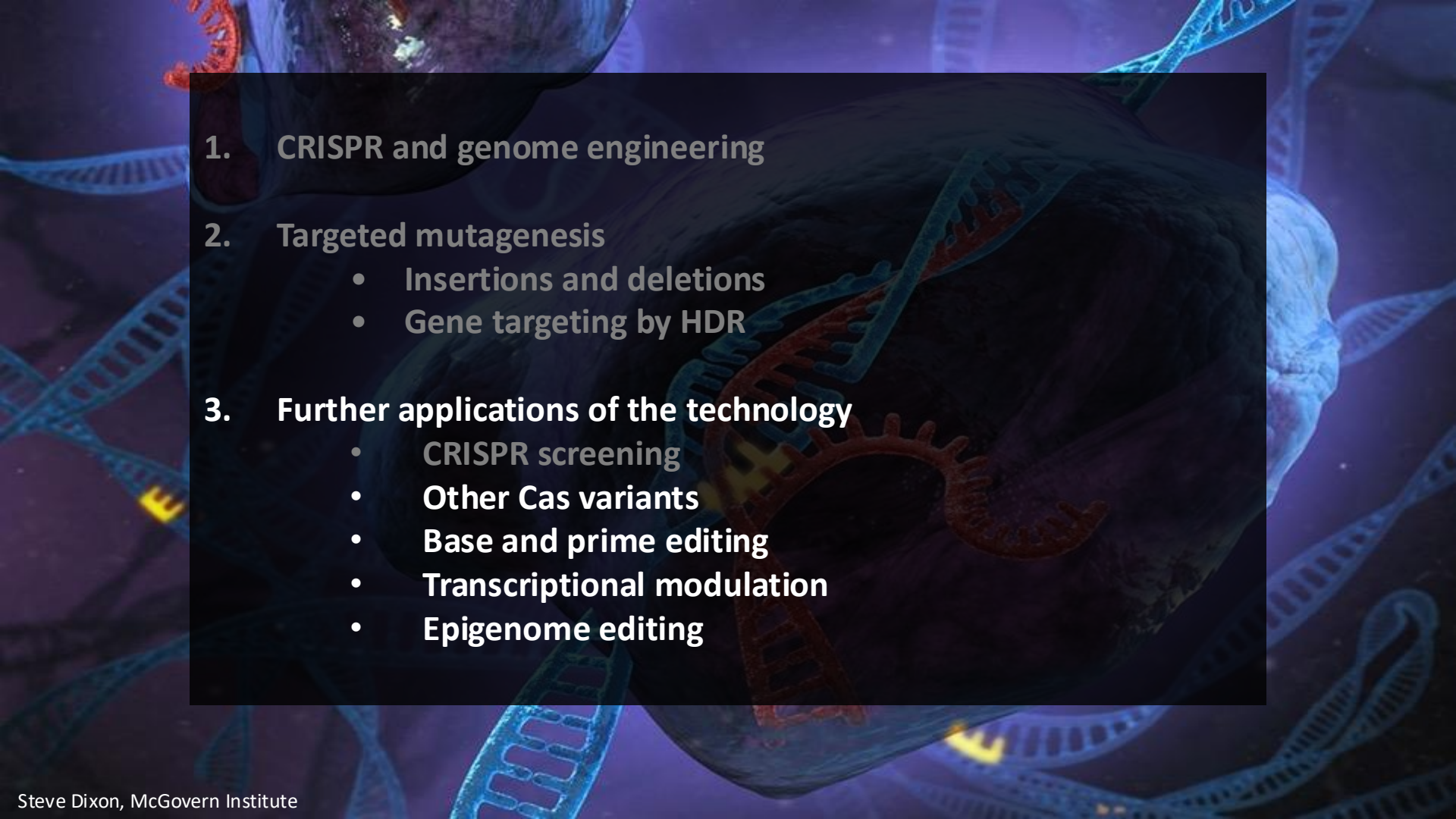


Single cell

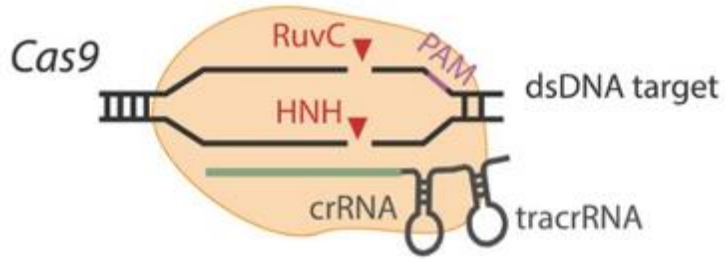


Imaging-based assays

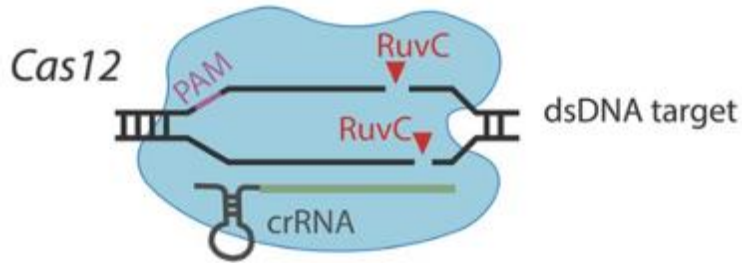


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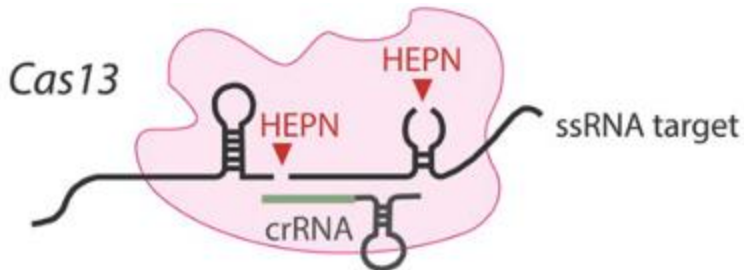
Other Cas proteins



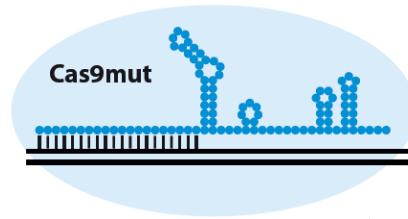
- Engineered PAM variants (e.g. NG)
- Alternative species (e.g. SaCas9)
- Enhanced specificity (e.g. HiFi Cas9, eSpCas9)



- Simple multiplexing (crRNA processing activity)
- Orthologous, independent system
- Collateral activity for diagnostics

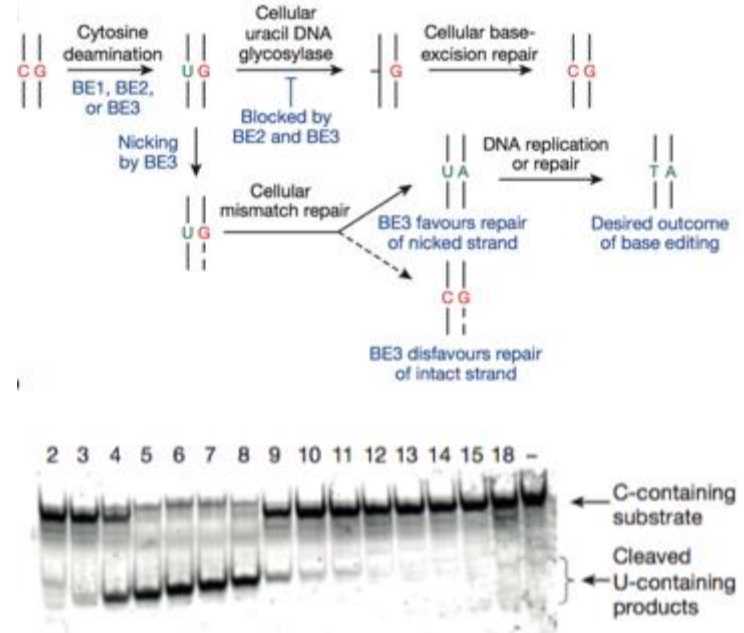
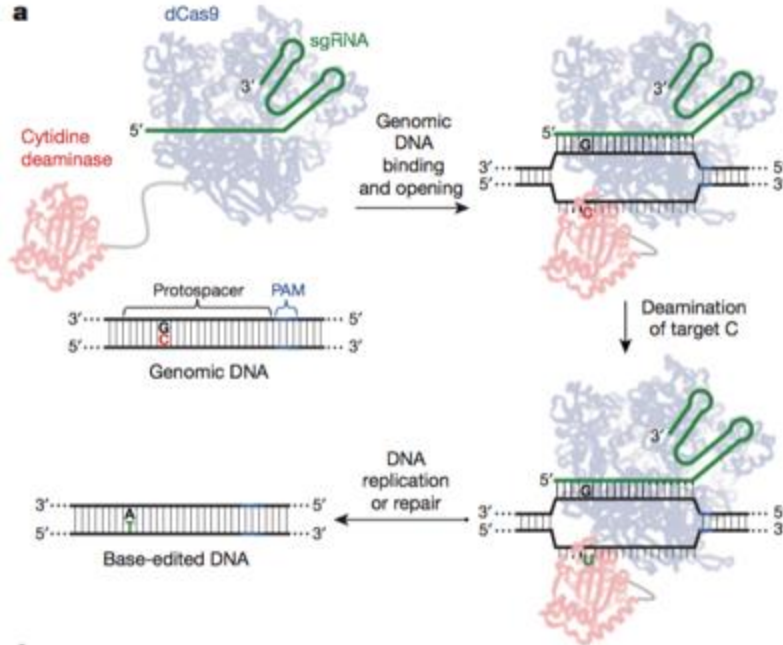


- RNA level silencing (transcript specific, non-coding RNAs)
- RNA modifying enzyme fusions



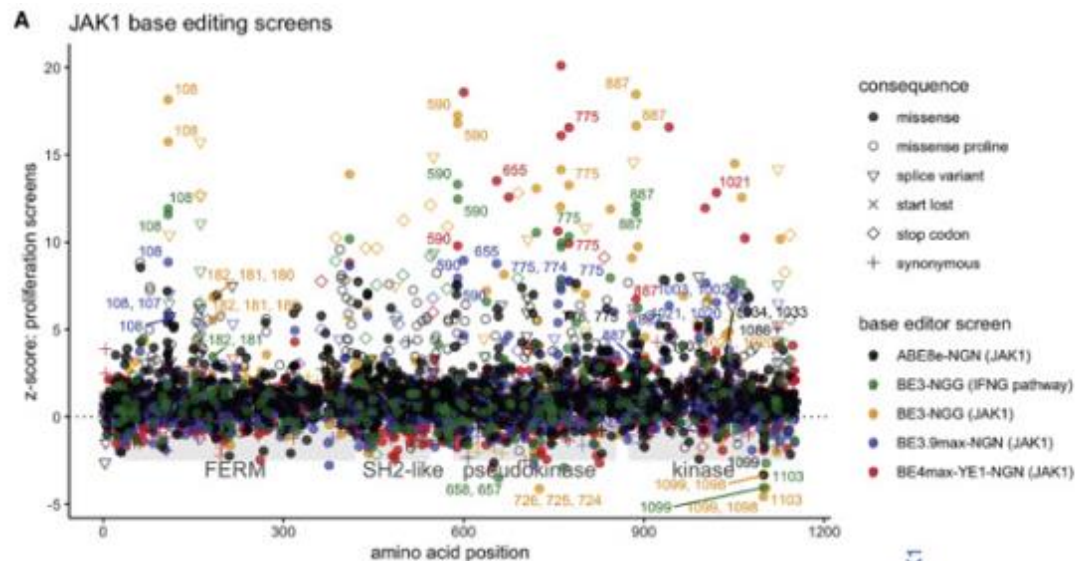
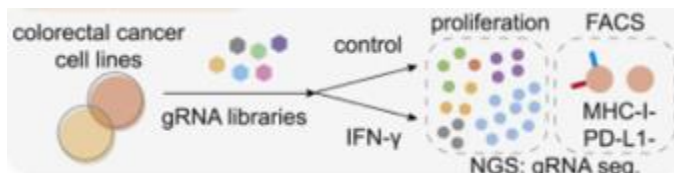
Sequence-specific DNA binding factor

Base Editing



- Targeting range limited by PAM (but variant e.g. Cas9-NG enzymes)
- Several nucleotides can be modified
- A-G editors also described (Gaudelli et al. Nature 2017, NatBiotech 2020)
- Also C-G editors (Kurt et al. NatBiotech 2020)

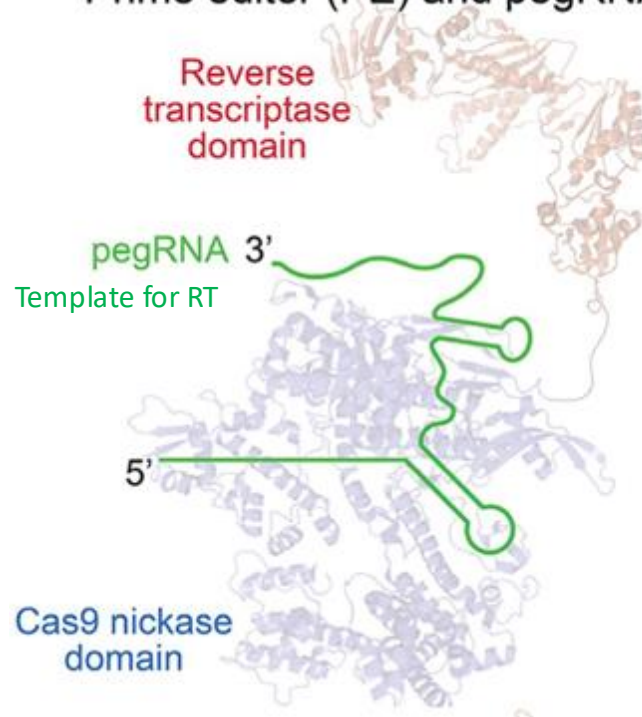
JAK1 tiling BE screen

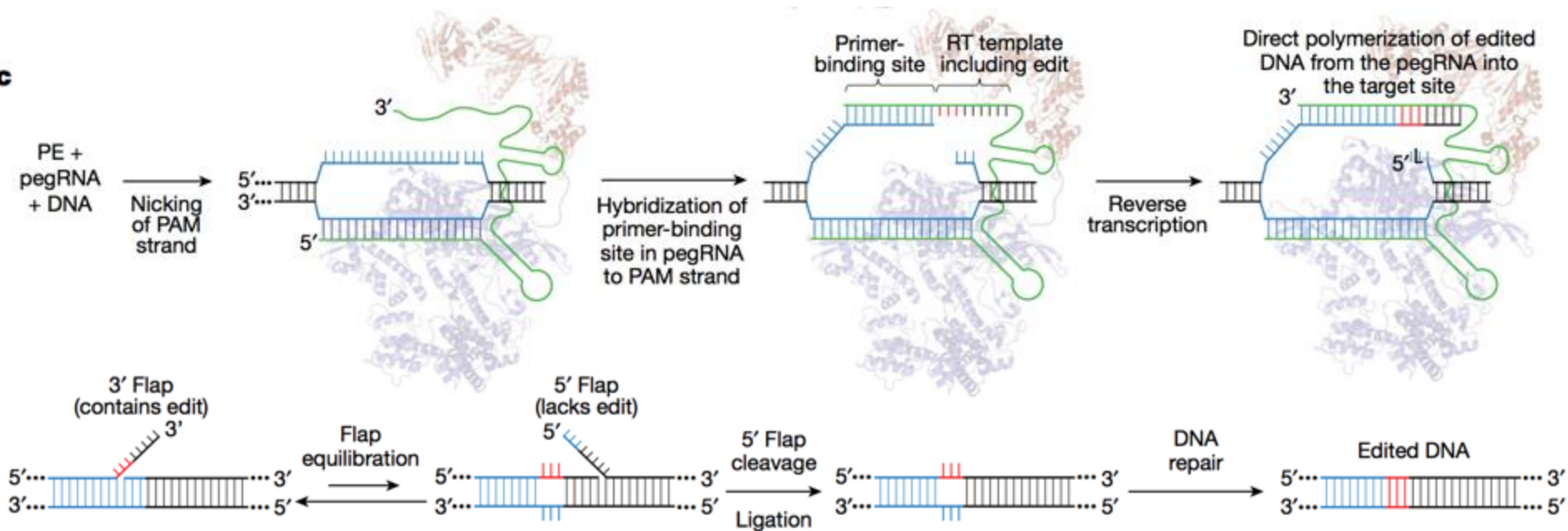


- Can screen across hundreds of thousands of variants at diverse sites
- Somewhat limited in targeting range
- Multiple edits within and sometimes outside window

Prime Editing

Prime editor (PE) and pegRNA



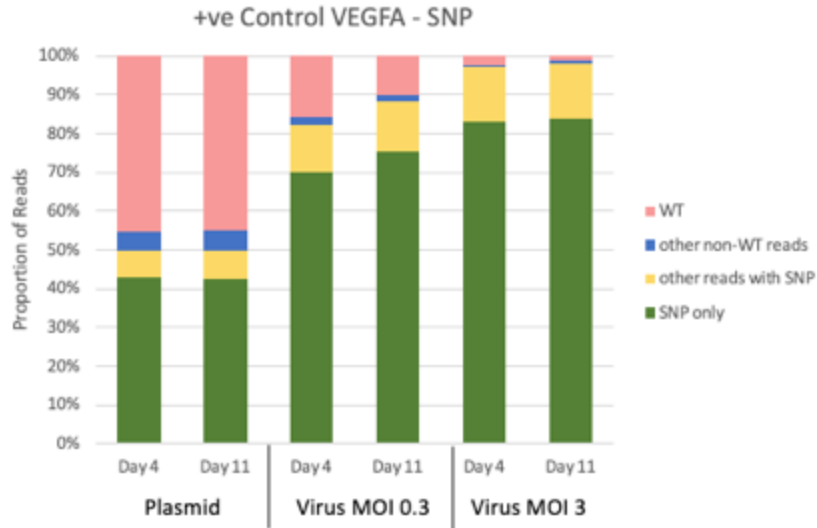
c

- Allows ANY base substitution in the genome... but efficiency still needs work

Prime Editing

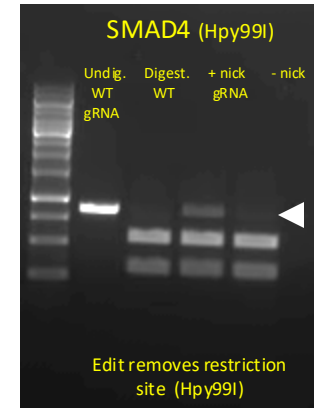
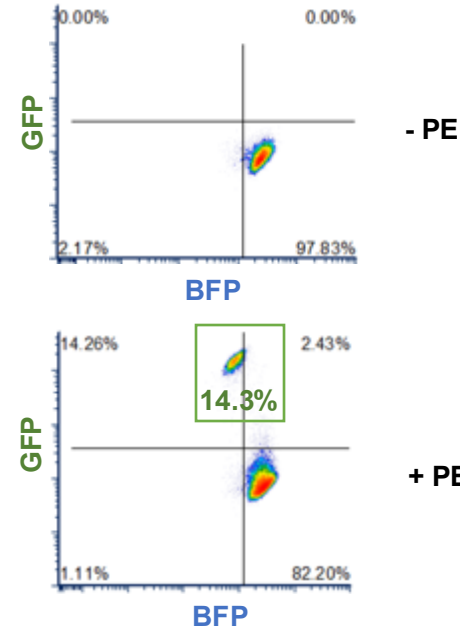
HAP1

Δ MLH1 (mismatch repair pathway)
PiggyBac inducible PE clonal line



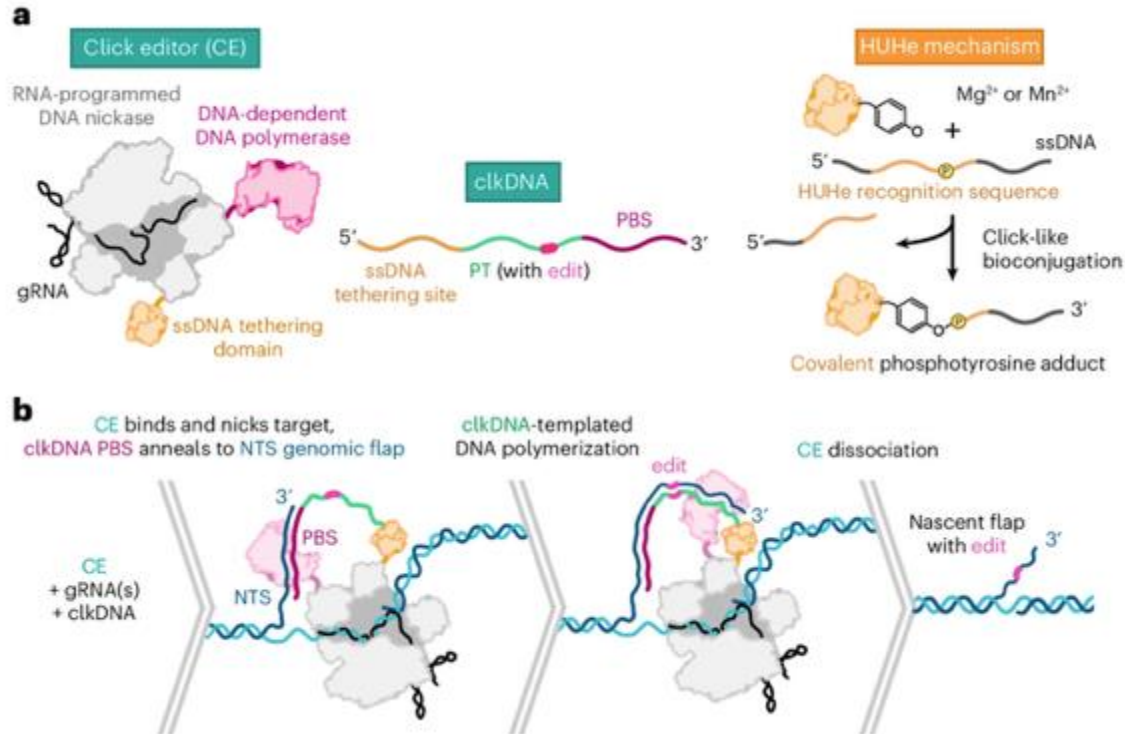
hiPSC

PE + dnMLH1 mRNA + sgRNA delivery

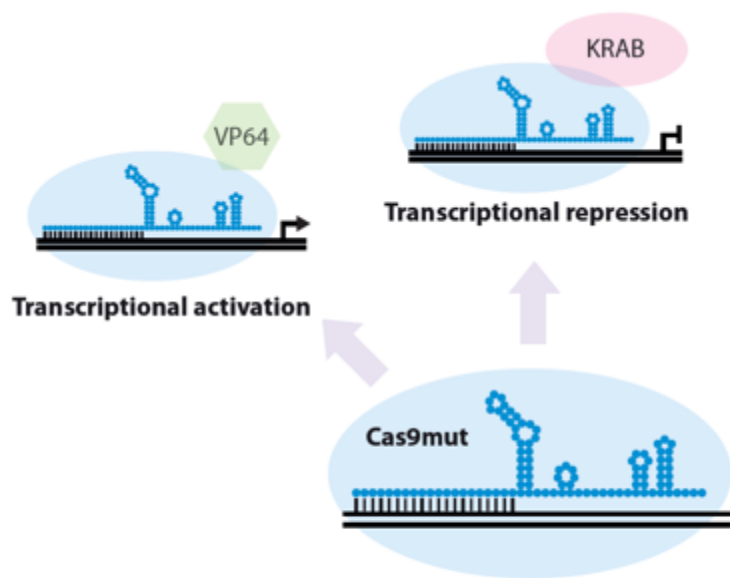


- Highly variable between sites and pegRNAs

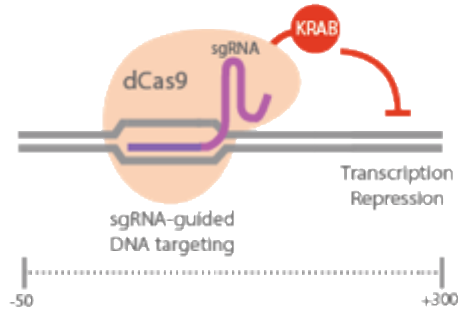
Click Editing



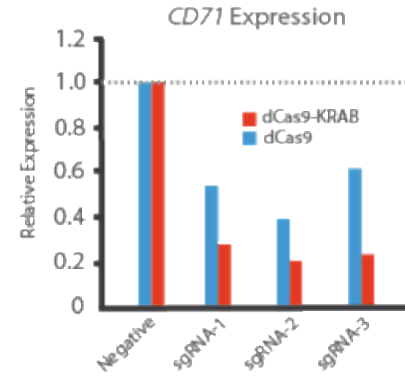
- Currently lower efficiency, but optimisation of proteins and DNA modifications will improve



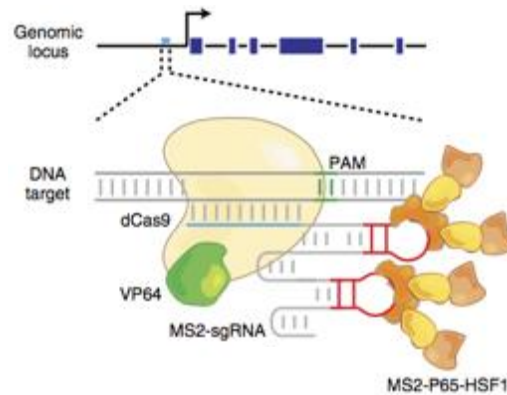
a



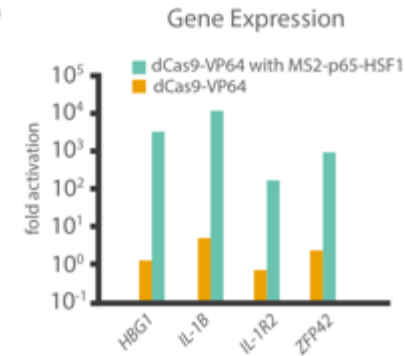
b



a

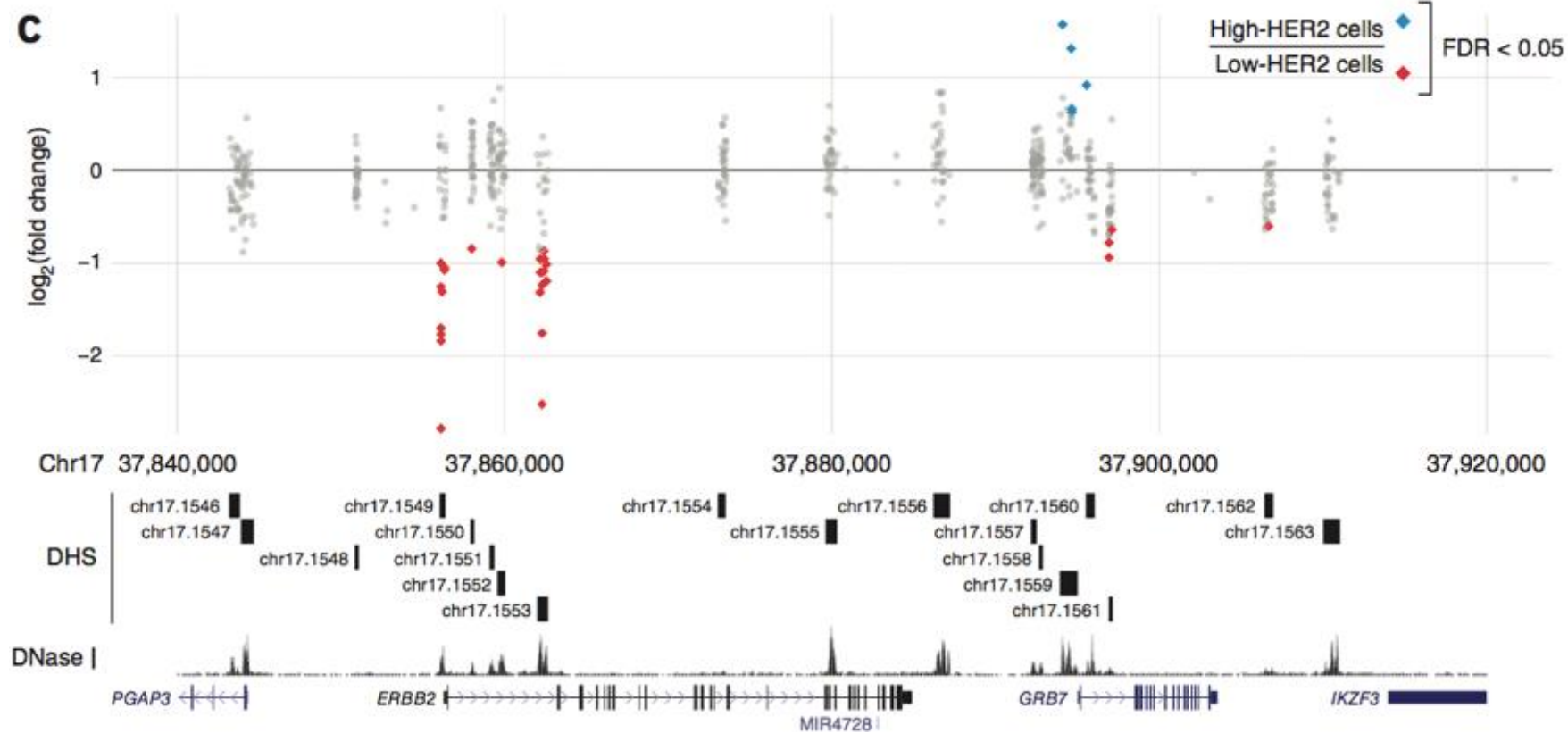


b

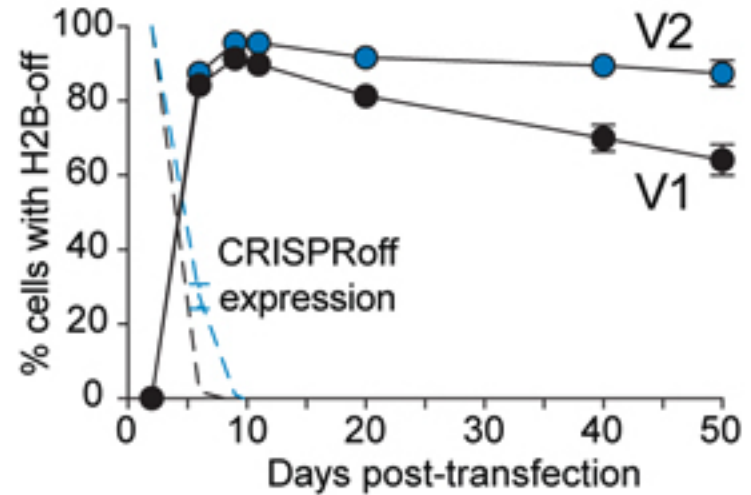
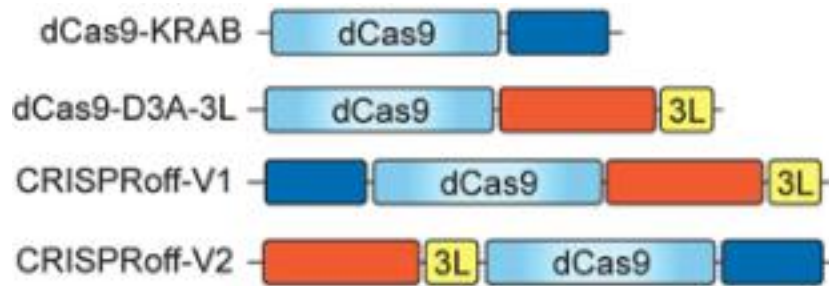


Genome-wide CRISPRi and CRISPRa libraries e.g. Gilbert et al. 2015, Konermann et al. 2015, Horlbeck et al. 2016, Weissman 2021

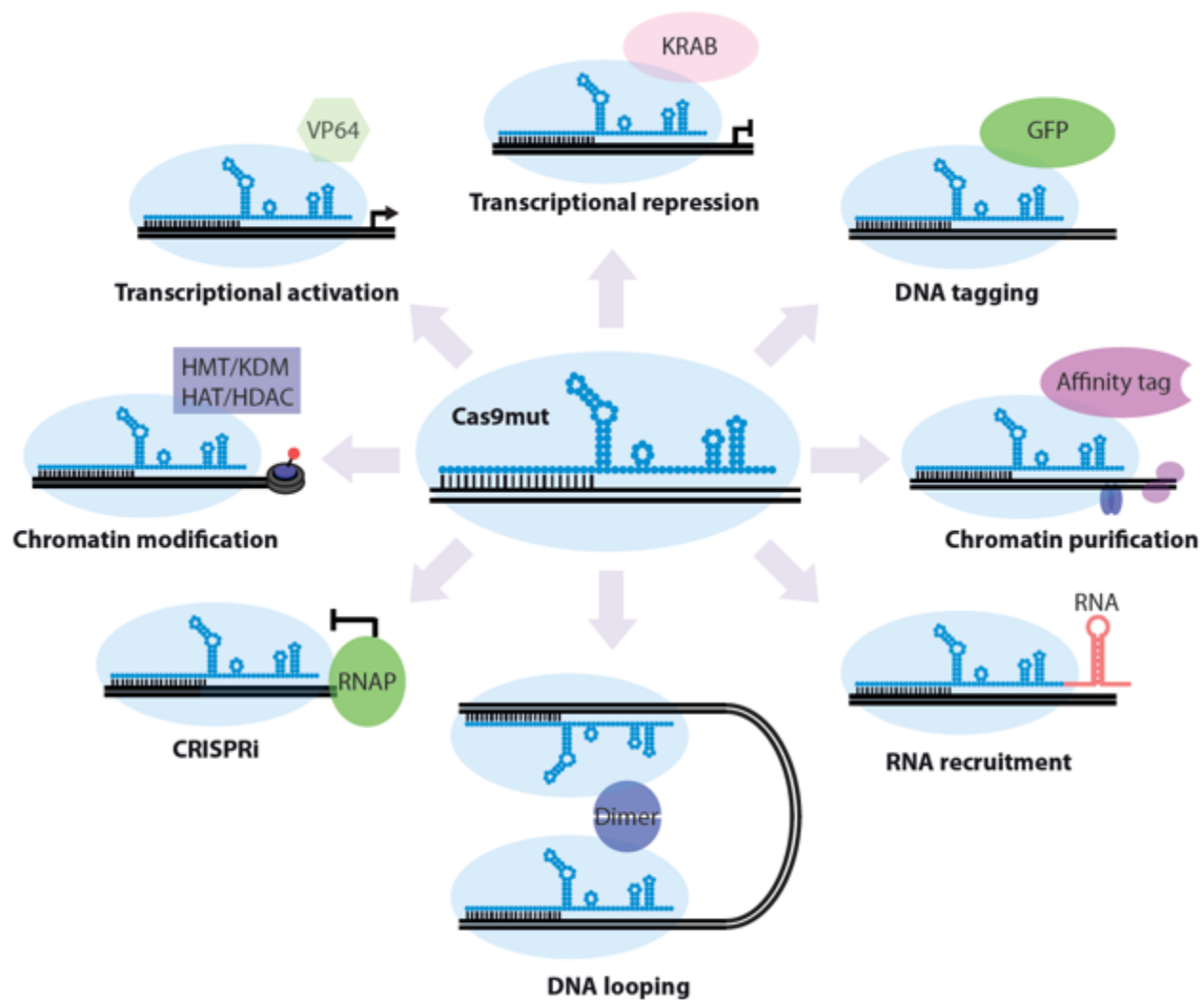
<https://www.addgene.org/pooled-library/>



CRISPR-off



- Fusion to KRAB, DNMT3A/3L causes long term epigenetic silencing





Henry Scowcroft
@oh_henry

If you genetically edit the lettuce genome, you can make it CRISPR
[#crisprfacts](#)

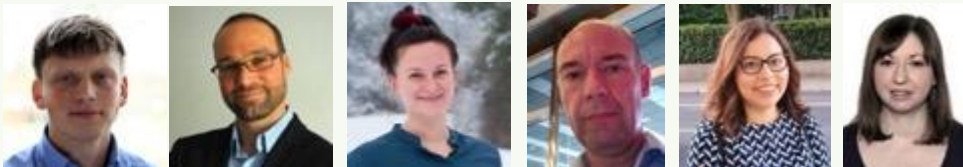


Andrew Maynard via Twitter

OpenTargets Neurodegeneration



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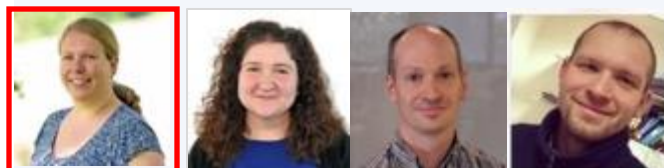
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Recent former members

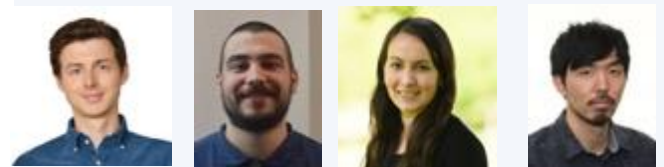


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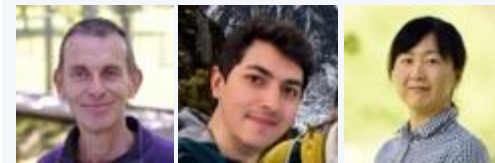
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